STEP BUY STEP KONFIGURASI SERVER DEBIAN 11 DI VIRTUALBOX

Digunakan untuk panduan belajar siswa-siswi SMKN 1 Sampang



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DAFTAR ISI

BAB I PENGENALAN

1. Opening .....
   1.1. Pengantar .....
   1.2. Pengenalan Linux .....

BAB II KONFIGURASI SERVER

2. Konfigurasi TCP/IP .....
   2.1. Topologi Jaringan .....
   2.2. Mengaktifkan Ethernet .....
   2.3. Konfigurasi IP Adress .....
   2.4. Menambahkan DNS .....
   2.5. Seting Repositori .....
      2.5.1. Instalasi Software VIA DVD/CD .....
      2.5.2. Repositori Via Jaringan .....
   2.6. Ip Address Alias .....
   2.7. Host Name .....
3. Remote Access .....
   3.1. Instalasi .....
   3.2. Konfigurasi .....
      3.2.1. Merubah port default ssh .....
      3.2.2. Modifikasi tampilan Login ssh .....
   3.3. Pengujian .....
      3.3.1. Remote Access via Localhost .....
      3.3.2. Remote Access via Windows .....
      3.3.3.

MODUL PRAKTIKUM
ADMINISTRASI SISTEM JARINGAN

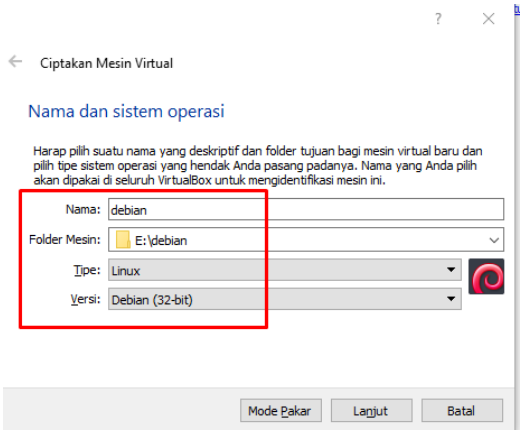
VIRTUAL BOX

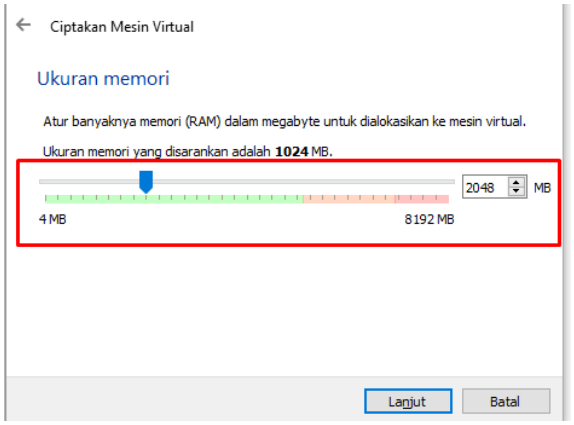
1. Membuat Project di Virtual Box

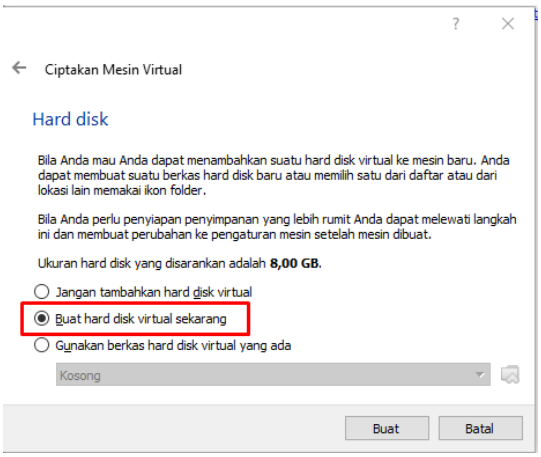
Ukuran Memori = 2048 MB → minimal 1024 MB
Ukuran HD = 20 GB → minimal 8 GB

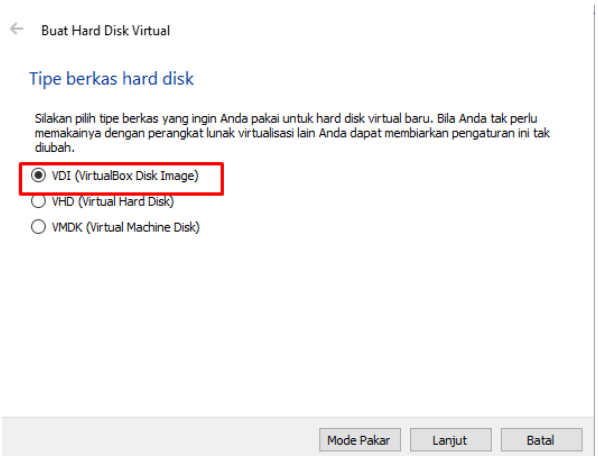


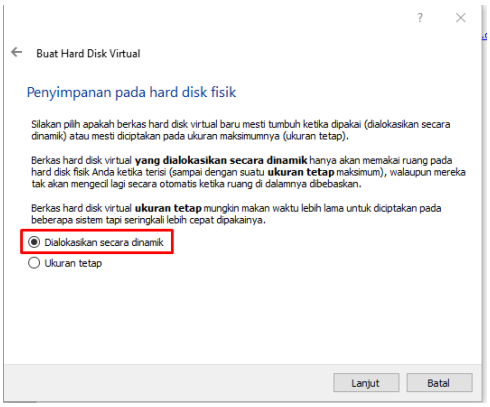
Lakukan seting seperti berikut

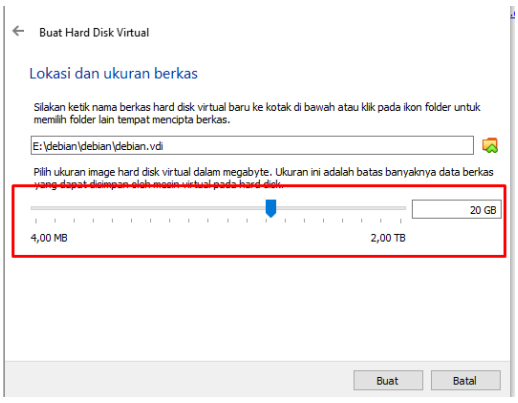




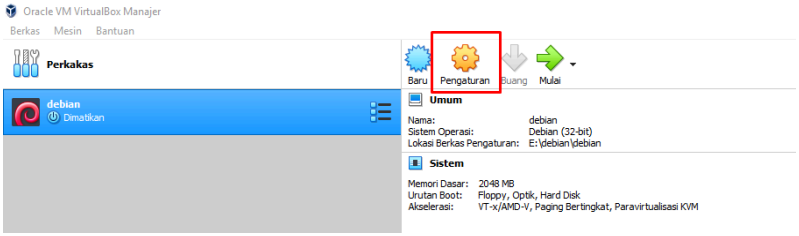


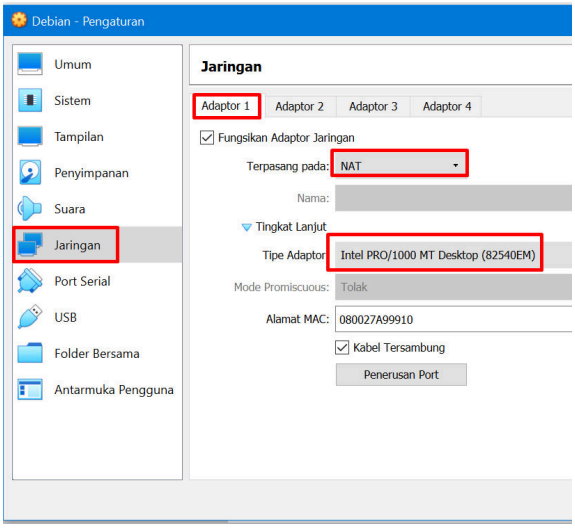


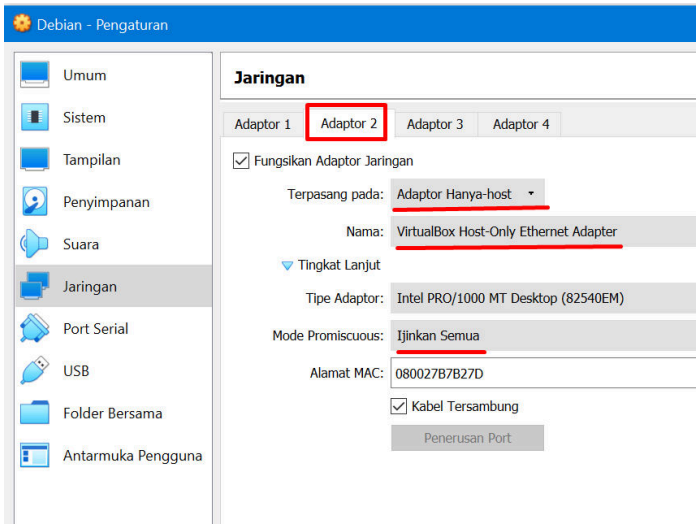




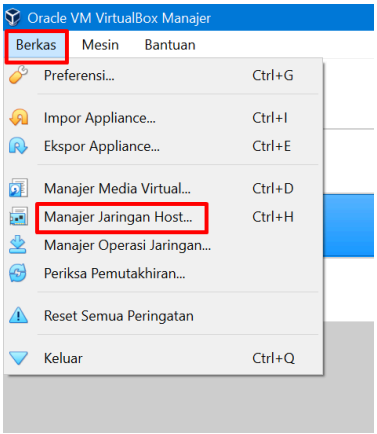
Jika selesai maka akan muncul di bagian kiri sebuah nama Virtual Project Anda
Lakukan **seting/pengaturan** lanjutan sebagaimana ilustrasi di bawah ini:

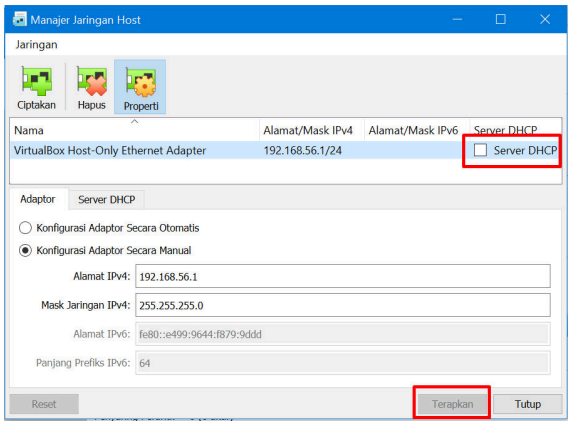






Pilihan Adaptor jaringan mengikuti ethernet masing-masing PC, **bedakan pilihan Adaptor 1 dan 2**
Matikan Opsi DHCP Server dari VirtualBox

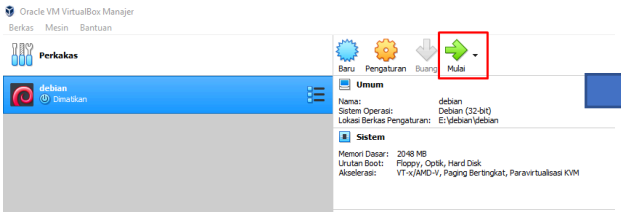


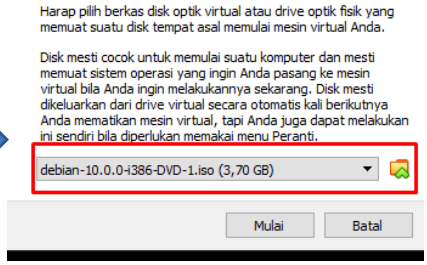


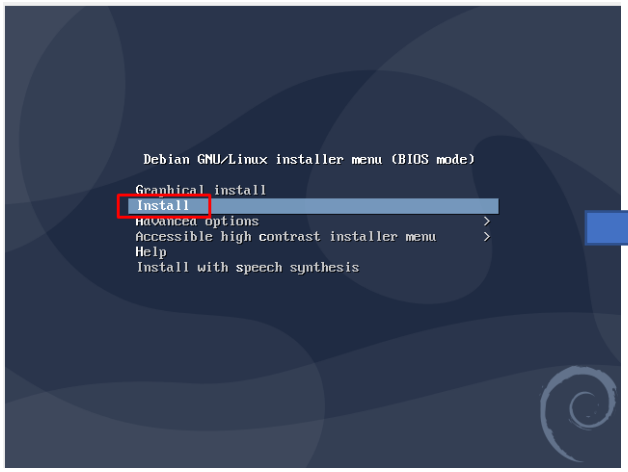
Jika sudah selesai, maka kita sudah bisa memulai ke langkah berikut nya, pastikan dalam tahap ini, kabel UTP/wifi laptop jangan di hubungkan terlebih dahulu

INSTALASI SISTEM OPERASI DEBIAN

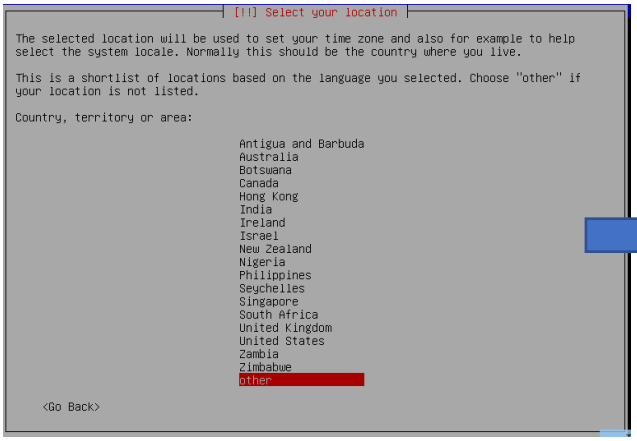
2. Instalasi Sistem Operasi Linux Debian 10

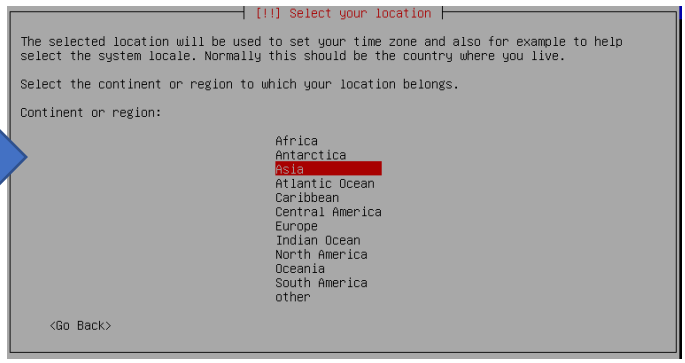


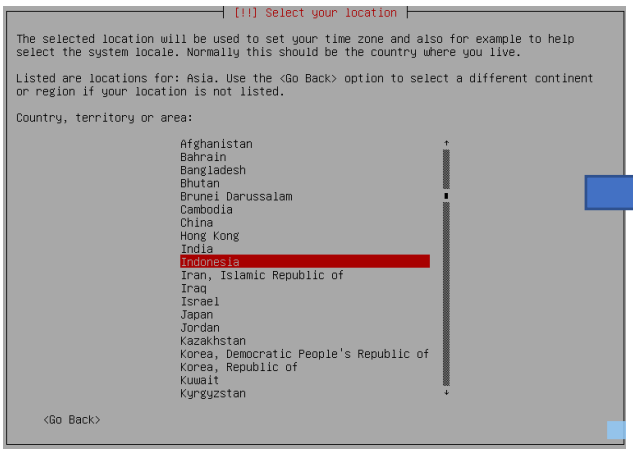




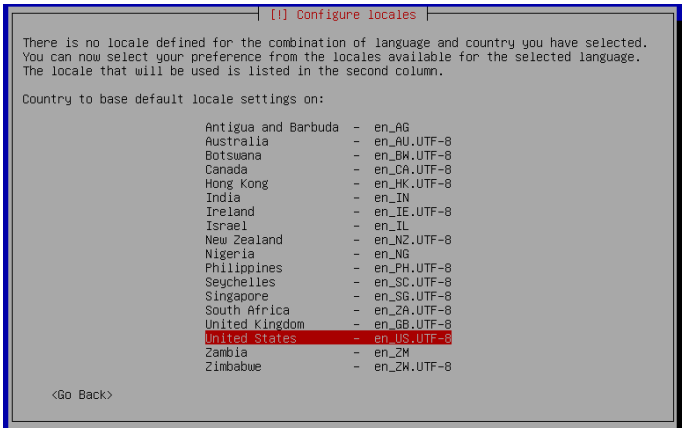




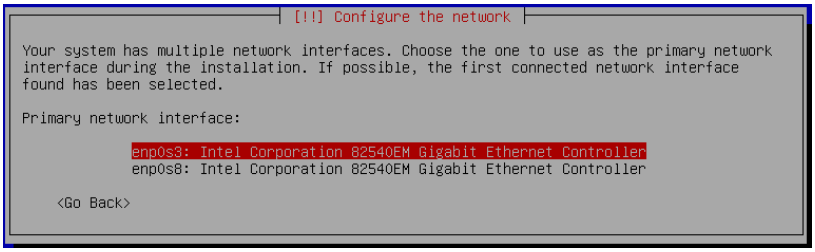


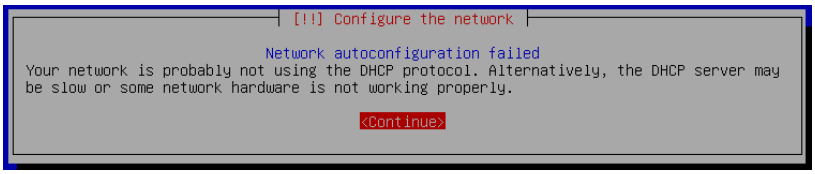




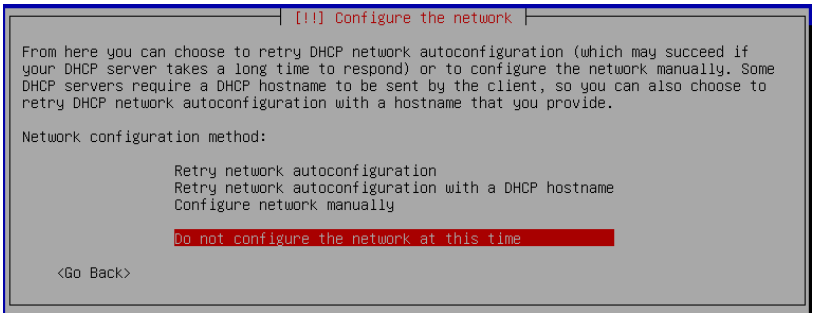


Pada tahap ini, terkadang masing-masing pc memiliki identifikasi yang berbeda untuk penamaan ethernet nya, pastikan anda catat dan ingat baik-baik

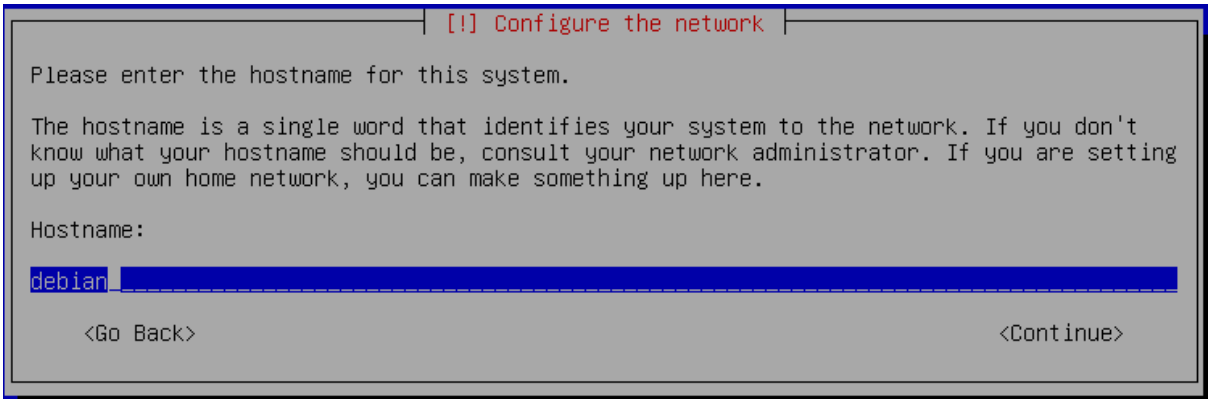




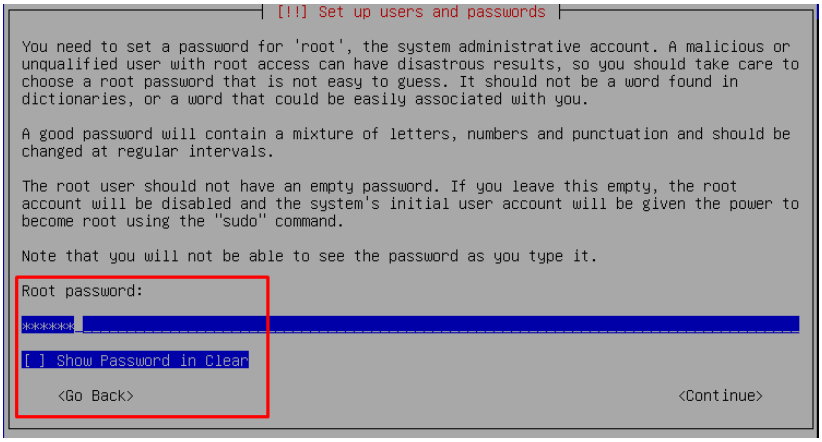
Jika di minta untuk melakukan konfigurasi jaringan, pilih saja **“Do not configure the network at this time”**
Karena semua konfigurasi akan kita lakukan nanti setelah selesai instalasi Debian



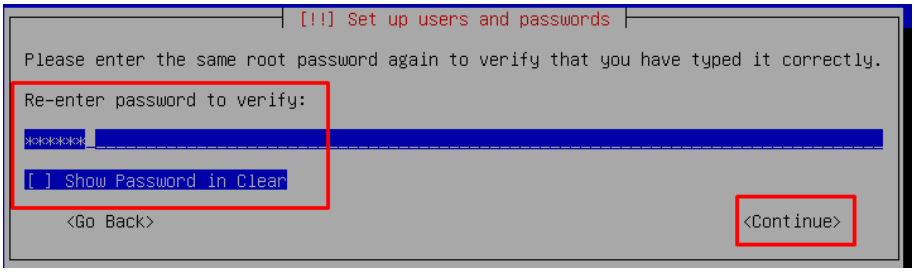
Biarkan saja nama host sesuai default nya (nanti akan kita konfigurasi kembali setelah selesai instalasi)



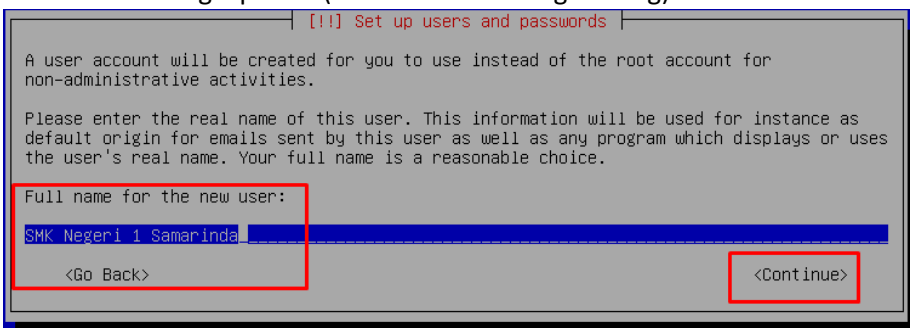
Isikan Password untuk user **“root”**



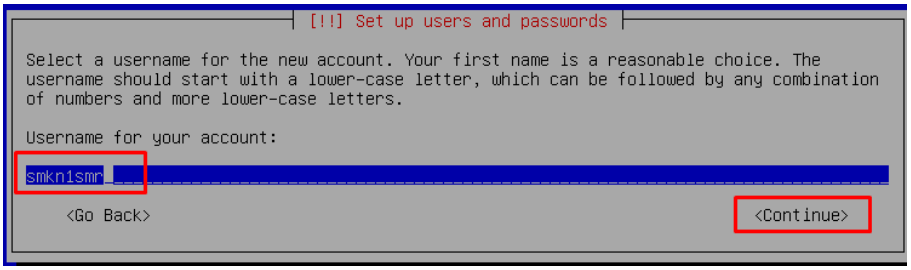
Ulangi kembali pengisian password (wajib sama dengan yang sebelumnya)



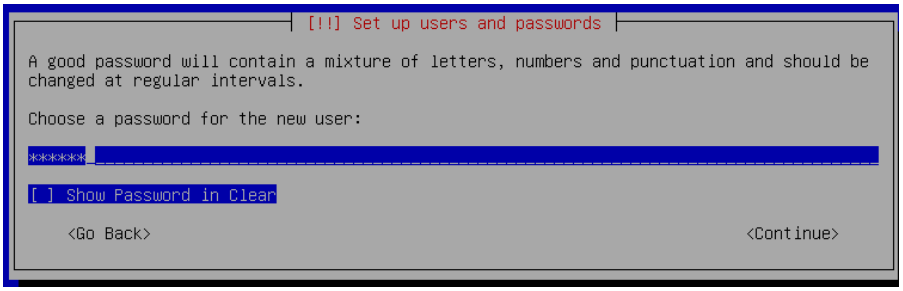
Isikan Nama Lengkap User (nama kalian masing-masing)



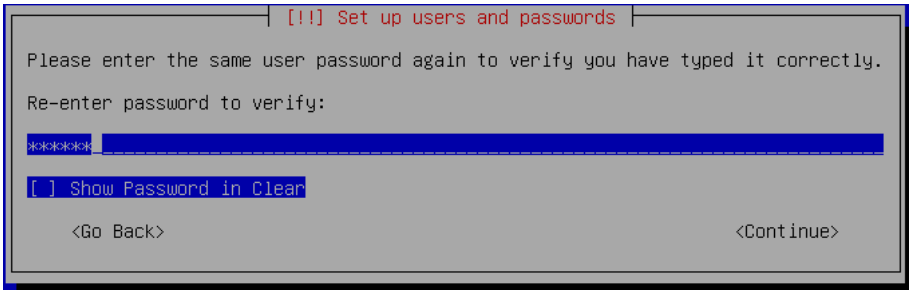
Isikan nama user



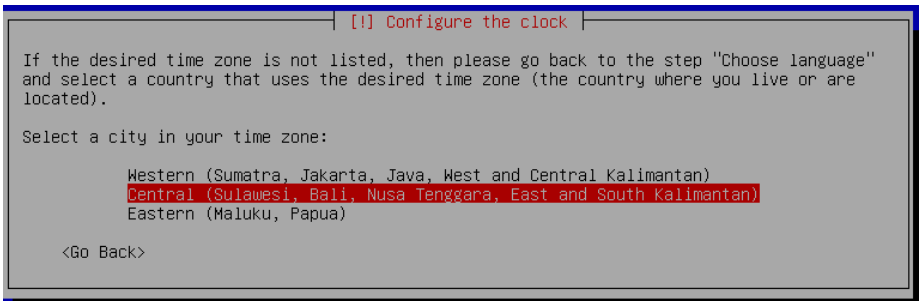
Isikan Password untuk user



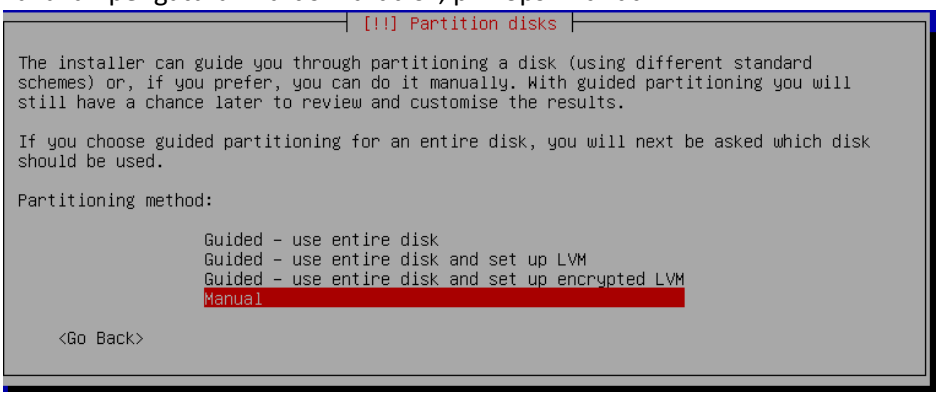
Ulangin kembali pengisian Password



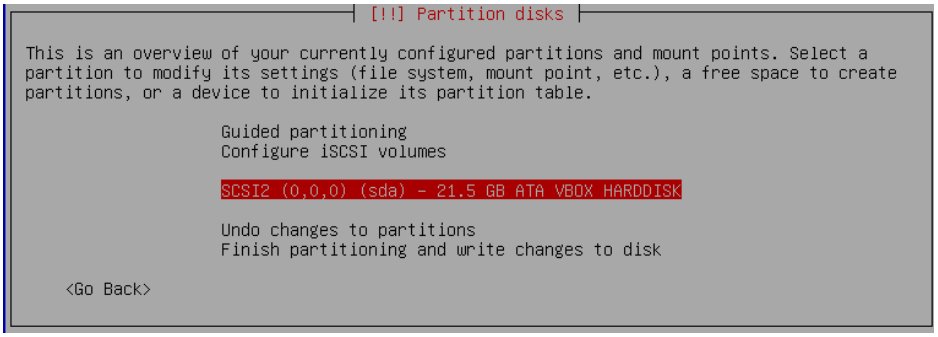
Sesuaikan pengaturan wilayah

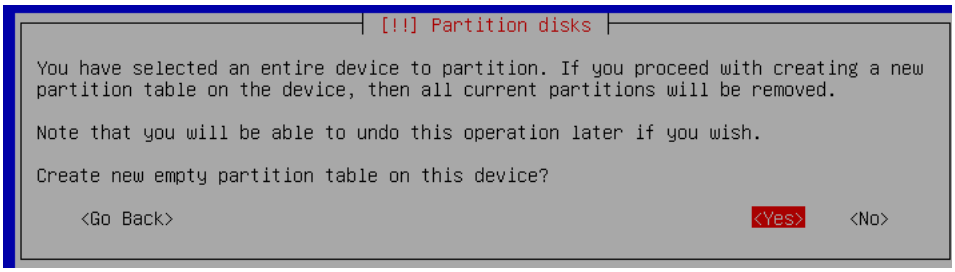


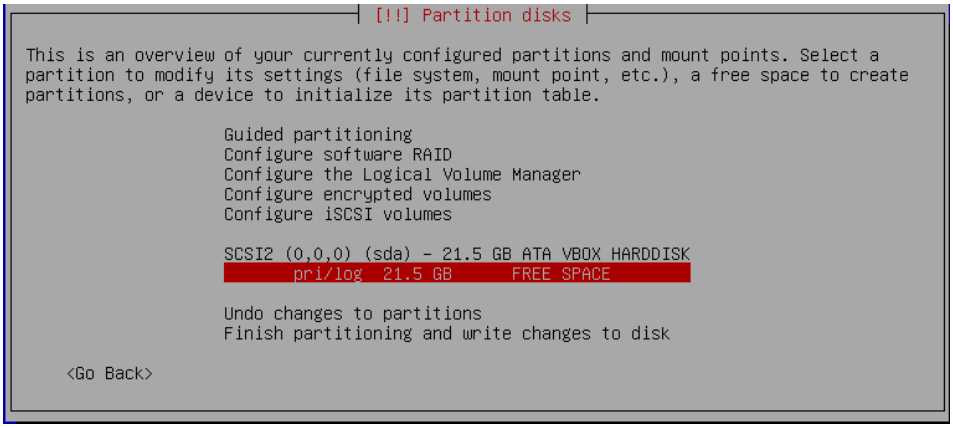
Lakukan pengaturan Partisi Harddisk, pilh opsi **Manual**



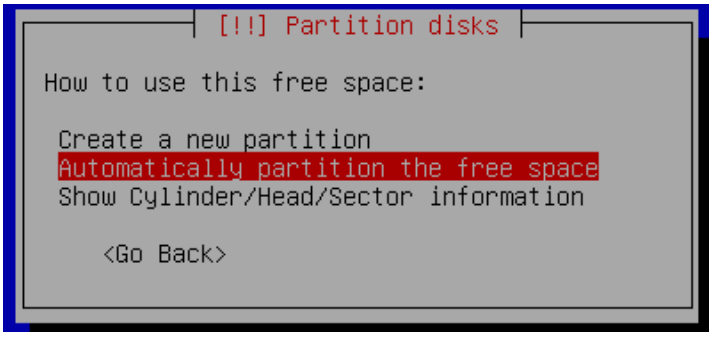
Pilih pada partisi **VBOX HARDDISK**

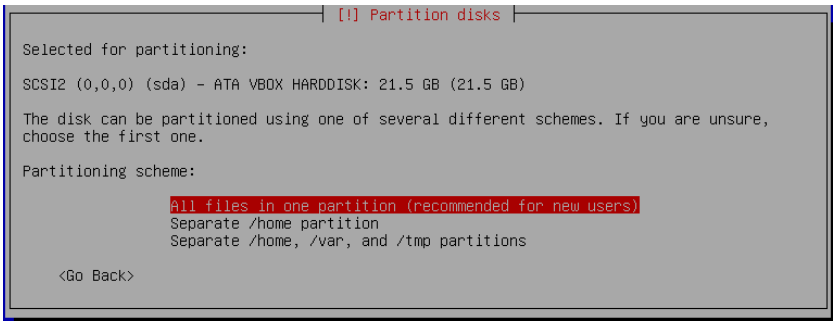


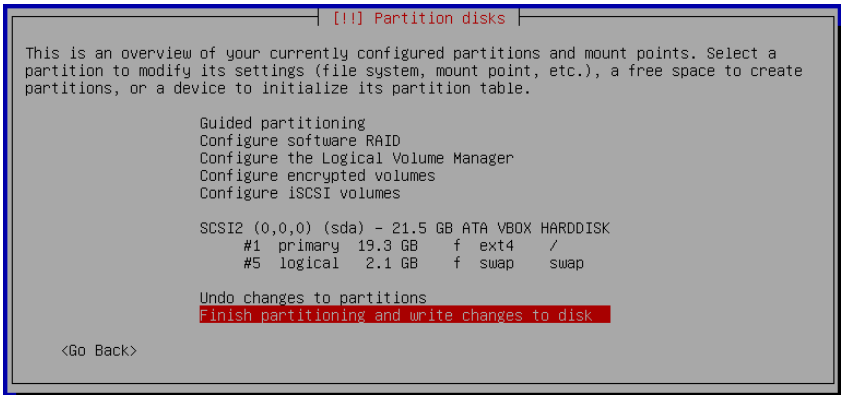




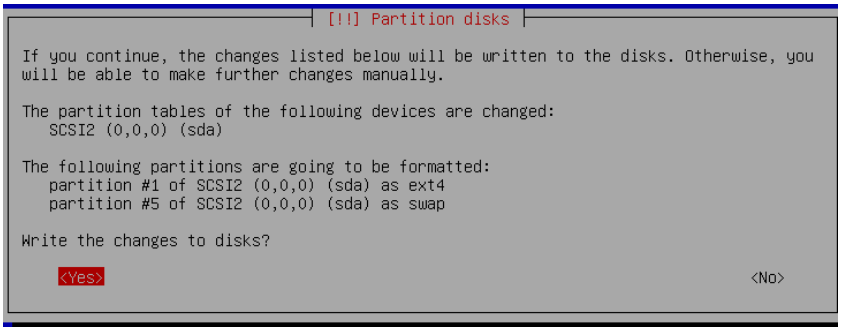
Untuk Praktikum kali ini, silahkan menggunakan opsi **Automatically partition the free space**



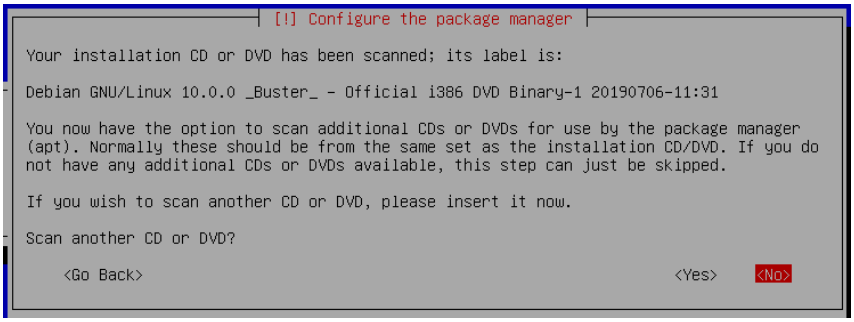




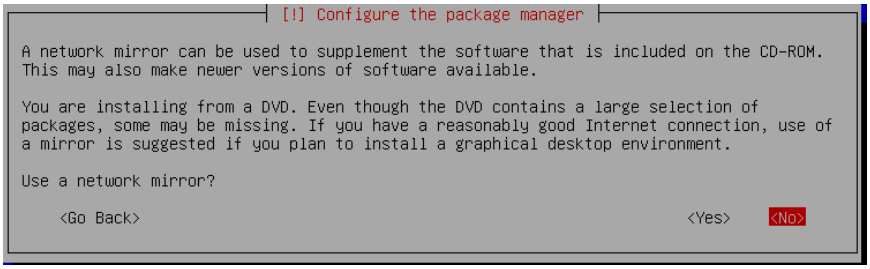
Pilih opsi **“yes”** agar sistem langsung menerapkan partisi yang telah kita konfigurasi



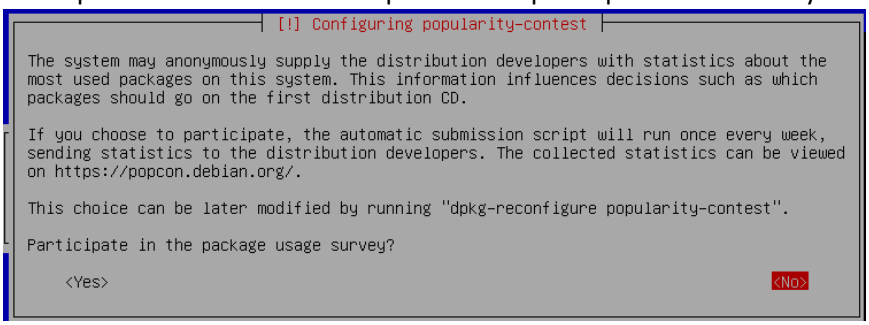
Pilih opsi **“no”,** nantinya opsi ini akan kita konfigurasi lebih lanjut di bagian **“sources.list”**



Pilih opsi **“no”**



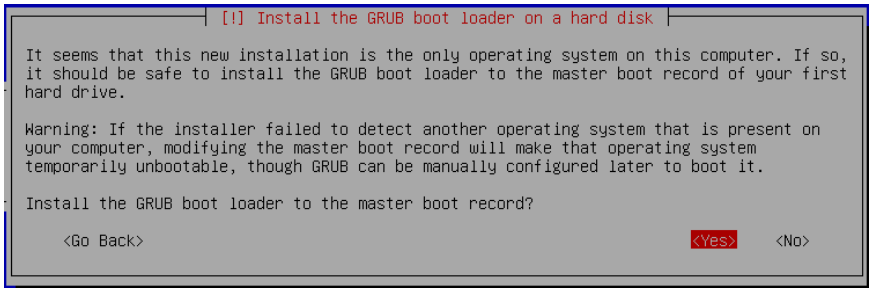
Pilih opsi **“no”** karena kita tidak perlu untuk partisipasi dalam survey

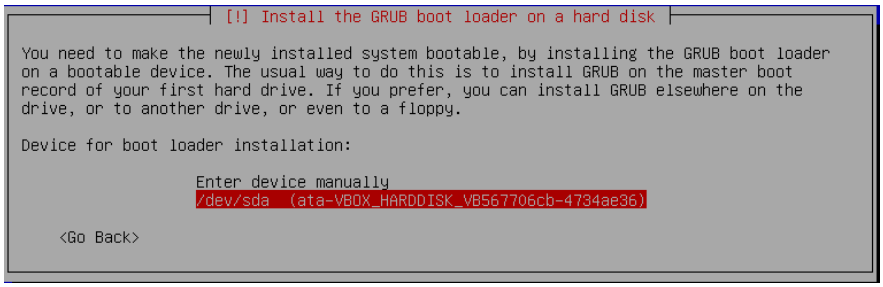


Pada pilihan aplikasi yang akan di install, hilangkan seluruh tanda * (bintang) menggunakan **“spasi”** sisakan satu saja pada bagian **Standard System Utilities**

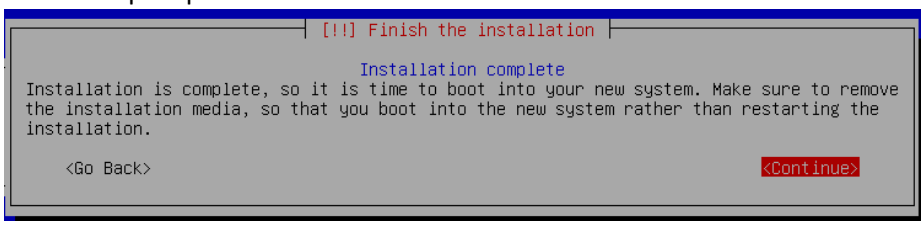


Di opsi berikut ini wajib memilih **Yes**, jika tidak Sistem Operasi Debian tidak dapat boot secara langsung





Pada Tahap ini proses instalasi debian 10 telah selesai



Selamat Anda telah menyelesaikan proses Instalasi Debian



Silahkan login menggunakan user yang telah di buat pada waktu proses instalasi

KONFIGURASI DASAR SERVER DEBIAN

3. Seting Interface Network

Pastikan Anda berada pada posisi **USER ROOT** untuk melakukan konfigurasi dengan perintah **su -** kemudian isikan password **root**

Cek Interface ethernet yang aktif dan terbaca pada debian Anda

ip a

```
root@rasyid-server:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp4s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
group default qlen 1000
    link/ether e8:de:27:01:20:15 brd ff:ff:ff:ff:ff:ff
    inet 192.168.110.2/24 brd 192.168.110.255 scope global enp4s0
       valid_lft forever preferred_lft forever
    inet6 fe80::eade:27ff:fe01:2015/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s31f6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default
 qlen 1000
    link/ether 6c:2b:59:d9:7f:18 brd ff:ff:ff:ff:ff:ff
root@rasyid-server:~#
```

Catat dan ingat baik-baik nama interface yang terdeteksi, tiap pc bisa menghasilkan penamaan yang berbeda.

Seting konfigurasi ip dengan aturan main sebagai berikut :

enp4s0	IP	:	192.168.110.xx	//ganti xx dengan ip kalian masing-masing.
	subnet	:	255.255.254.0	
	Gateway	:	192.168.110.1	

pico /etc/network/interfaces

```
root@smkn1smr:/# pico /etc/network/interfaces
```

Tampilan awal

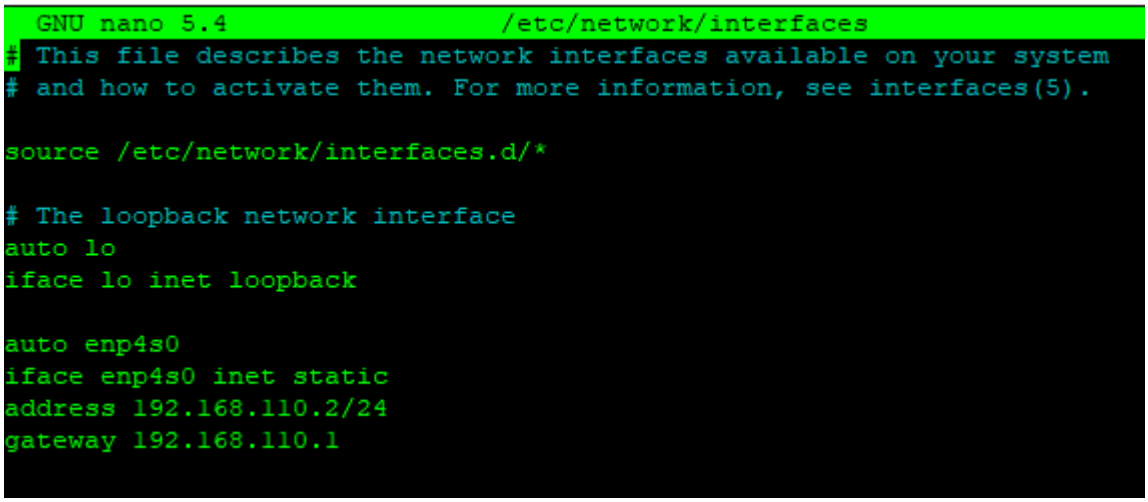
```
  GNU nano 3.2                          /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback
```

Tampilan setelah di konfigurasi



Sesuaikan IP nya ya

Simpan setingan Anda dengan cara **ctrl + o** (huruf o bukan huruf nol)

Keluar aplikasi **ctrl + x**

Lakukan restart service networking agar konfigurasi yang di buat langsung dapat di gunakan

/etc/init.d/networking restart

atau

systemctl restart networking

Silahkan gunakan salah satu saja

```
root@smkn1smr:/# /etc/init.d/networking restart
[ ok ] Restarting networking (via systemctl): networking.service.
root@smkn1smr:/# _
```

Cek kembali apakah setingan Anda sudah berhasil dengan perintah **ip a**

```
root@rasyid-server:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp4s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default
 qlen 1000
    link/ether e8:de:27:01:20:15 brd ff:ff:ff:ff:ff:ff
    inet 192.168.110.2/24 brd 192.168.110.255 scope global enp4s0
       valid_lft forever preferred_lft forever
    inet6 fe80::eade:27ff:fe01:2015/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s31f6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 6c:2b:59:d9:7f:18 brd ff:ff:ff:ff:ff:ff
root@rasyid-server:~#
```

Jika terdapat eror perikasa kembali konfigurasi yang telah Anda buat

- **Pengujian**

  Tes ke jaringan lokal yang tersedia

  # ping 192.168.110.1 (gateway lokal)

  # ping 192.168.110.10 (repo lokal)

  Jika sukses seperti ini tampilannya

```
root@rasyid-server:~# ping 192.168.110.1
PING 192.168.110.1 (192.168.110.1) 56(84) bytes of data.
64 bytes from 192.168.110.1: icmp_seq=1 ttl=64 time=0.578 ms
64 bytes from 192.168.110.1: icmp_seq=2 ttl=64 time=0.291 ms
64 bytes from 192.168.110.1: icmp_seq=3 ttl=64 time=0.304 ms
^C
--- 192.168.110.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2055ms
rtt min/avg/max/mdev = 0.291/0.391/0.578/0.132 ms
root@rasyid-server:~# ping 192.168.110.3
PING 192.168.110.3 (192.168.110.3) 56(84) bytes of data.
64 bytes from 192.168.110.3: icmp_seq=1 ttl=128 time=0.908 ms
64 bytes from 192.168.110.3: icmp_seq=2 ttl=128 time=0.548 ms
64 bytes from 192.168.110.3: icmp_seq=3 ttl=128 time=0.536 ms
^C
--- 192.168.110.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 0.536/0.664/0.908/0.172 ms
root@rasyid-server:~#
```

Note :

Mengaktifkan dan menonaktifkan ethernet gunakan perintah

ip link set dev 'device' up
ip link set dev 'device' down
Jika masih terbiasa dengan perintah ifconfig, maka wajib melakukan instalasi paket **net-tools**
apt-get install net-tools
ifup 'device'
ifdown 'device'
Ganti 'device' dengan nama ether misal enp0s3 tanpa petik

SETING FILE RESOLV.CONF

4. Konfigurasi IP DNS secara manual

Konfigurasi berikut tersimpan di dalam file **resolv.conf** untuk mengarahkan server ke dns lokal dan ke dns publik (agar dapat terkoneksi ke internet)

pico /etc/resolv.conf

```
root@smkn1smr:/# pico /etc/resolv.conf
```

Umumnya file tersebut belom berisi konfigurasi sama sekali, isikan konfigurasi **resolv.conf** seperti berikut :
Isikan nama domain dan ip Anda
Domain kalian = nama.sch.id → rasyid.sch.id
Name server = ip dari ethernet kalian yang diseting pada ip address

```
  GNU nano 5.4                                  /etc/resolv.conf
nameserver 192.168.110.1
search rasyid.sch.id
nameserver 192.168.110.2
nameserver 8.8.8.8
nameserver 8.8.4.4
```

- **Pengujian**

  Lalu tes dengan ping ke domain publik, misal detik.com

  # ping detik.com

```
root@debian:~# ping detik.com
PING detik.com (103.49.221.211) 56(84) bytes of data.
64 bytes from 103.49.221.211: icmp_seq=1 ttl=56 time=34.5 ms
64 bytes from 103.49.221.211: icmp_seq=2 ttl=56 time=32.1 ms
64 bytes from 103.49.221.211: icmp_seq=3 ttl=56 time=31.7 ms
64 bytes from 103.49.221.211: icmp_seq=4 ttl=56 time=31.6 ms
64 bytes from 103.49.221.211: icmp_seq=5 ttl=56 time=33.1 ms
64 bytes from 103.49.221.211: icmp_seq=6 ttl=56 time=32.3 ms
64 bytes from 103.49.221.211: icmp_seq=7 ttl=56 time=32.3 ms
^C
```

KONFIGURASI HOST DAN HOSTNAME

5. Mengubah nama host dan hostname

Konfigurasi Hosts

pico /etc/hosts

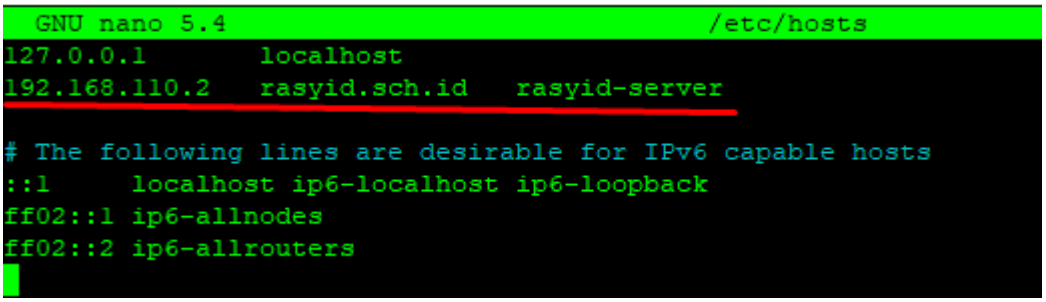
```
root@smkn1smr:/# pico /etc/hosts
```

```
  GNU nano 3.2                                  /etc/hosts

127.0.0.1       localhost
127.0.1.1       debian

# The following lines are desirable for IPv6 capable hosts
::1     localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

Ubah menjadi


GNU nano 5.4 /etc/hosts
127.0.0.1 localhost
192.168.110.2 rasyid.sch.id rasyid-server
The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

Konfigurasi Hostname

pico /etc/hostname

```
root@smkn1smr:/# pico /etc/hostname
```

```
GNU nano 3.2
debian
```

ubah menjadi

```
GNU nano 3.2
rasyid-server
```

KONFIGURASI REPOSITORI KE SERVER LOKAL

6. Konfigurasi Repositori

Hubungkan debian dengan sources lokal di server lokal Laptop kalian

pico /etc/apt/sources.list

```
root@smkn1smr:/# pico /etc/apt/sources.list
```

```
GNU nano 3.2                    /etc/apt/sources.list
#
# deb cdrom:[Debian GNU/Linux 10.0.0 _Buster_ - Official i386 DVD Binary-1 20190706-11:31]/ buster $
deb cdrom:[Debian GNU/Linux 10.0.0 _Buster_ - Official i386 DVD Binary-1 20190706-11:31]/ buster co$
# Line commented out by installer because it failed to verify:
#deb http://security.debian.org/debian-security buster/updates main contrib
# Line commented out by installer because it failed to verify:
#deb-src http://security.debian.org/debian-security buster/updates main contrib
# buster-updates, previously known as 'volatile'
# A network mirror was not selected during install.  The following entries
# are provided as examples, but you should amend them as appropriate
# for your mirror of choice.
#
# deb http://deb.debian.org/debian/ buster-updates main contrib
# deb-src http://deb.debian.org/debian/ buster-updates main contrib
```

Ubah menjadi

```
GNU nano 5.4                                /etc/apt/sources.list
# deb cdrom:[Debian GNU/Linux 11.0.0 _Bullseye_ - Official amd64 DVD Binary-1 20210814-10:04]/ bullseye contrib main

#deb cdrom:[Debian GNU/Linux 11.0.0 _Bullseye_ - Official amd64 DVD Binary-1 20210814-10:04]/ bullseye contrib main
deb [trusted=yes] http://192.168.110.10/debian/ bullseye main
deb-src [trusted=yes] http://192.168.110.10/debian/ bullseye main

# Line commented out by installer because it failed to verify:
#deb http://security.debian.org/debian-security bullseye-security main contrib
# Line commented out by installer because it failed to verify:
#deb-src http://security.debian.org/debian-security bullseye-security main contrib

# bullseye-updates, to get updates before a point release is made;
# see https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#_updates_and_backports
# A network mirror was not selected during install.  The following entries
# are provided as examples, but you should amend them as appropriate
# for your mirror of choice.
#
# deb http://deb.debian.org/debian/ bullseye-updates main contrib
# deb-src http://deb.debian.org/debian/ bullseye-updates main contrib
```

Berikan tanda **#** pada seluruh isi file sources.list lalu tambahkan baris perintah berikut di bagian bawah

deb [trusted=yes] http://192.168.110.10/debian/ bullseye main

deb-src [trusted=yes] http://192.168.110.10/debian/ bullseye main

Jika ingin terhubung dengan repository online yang tersedia di Indonesia dapat menggunakan alamat berikut ini:

Data Utama Surabaya

deb http://kartolo.sby.datautama.net.id/debian/ buster main contrib non-free

deb http://kartolo.sby.datautama.net.id/debian/ buster-updates main contrib non-free

deb http://kartolo.sby.datautama.net.id/debian-security/ buster/updates main contrib non-free

Kambing UI

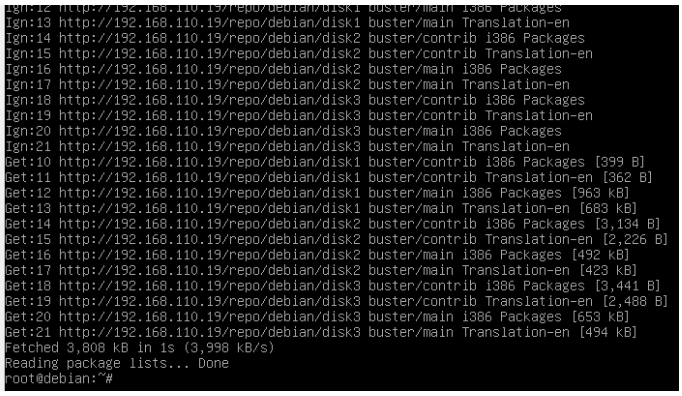
deb http://kambing.ui.ac.id/debian/ buster main contrib non-free
deb http://kambing.ui.ac.id/debian/ buster-updates main contrib non-free
deb http://kambing.ui.ac.id/debian-security/ buster/updates main contrib non-free

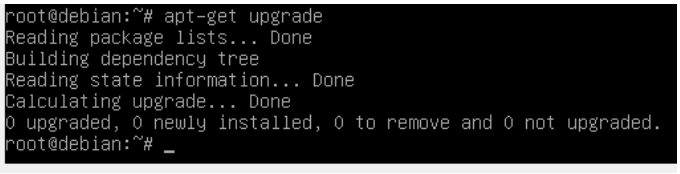
Kebo VLSM
deb http://kebo.vlsm.org/debian/ buster main contrib non-free
deb http://kebo.vlsm.org/debian/ buster-updates main contrib non-free
deb http://kebo.vlsm.org/debian-security/ buster/updates main contrib non-free

Mirror UNEJ
deb http://mirror.unej.ac.id/debian/ buster main contrib non-free
deb http://mirror.unej.ac.id/debian/ buster-updates main contrib non-free
deb http://mirror.unej.ac.id/debian-security/ buster/updates main contrib non-free

Silahkan gunakan salah satunya yang menurut kalian paling stabil

- **Pengujian**
  Lakukan update dan upgrade pada sources.list
  # apt-get update
  # apt-get upgrade





REMOTE ACCESS

7. **Install Aplikasi**
   # apt-get install openssh-server

INSTALASI DAN KONFIGURASI DOMAIN LOKAL

8. Konfigurasi DNS

Install aplikasi utama untuk membuat DNS

apt-get install bind9

```
root@debian:~# apt-get install bind9
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  bind9utils dns-root-data net-tools python3-ply
Suggested packages:
  bind9-doc dnsutils resolvconf ufw python-ply-doc
The following NEW packages will be installed:
  bind9 bind9utils dns-root-data net-tools python3-ply
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,390 kB of archives.
After this operation, 5,324 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Lakukan konfigurasi bind seperti berikut ini

pico /etc/bind/named.conf.local

```
 GNU nano 5.4                          /etc/bind/named.conf.local
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "rasyid.sch.id" {
        file "/etc/bind/rasyid";
        type master;
};
zone "110.168.192.in-addr.arpa" {
        file "/etc/bind/db.192";
        type master;
};
```

```
zone “rasyid.sch.id” {                    //Ganti rasyid.sch.id sesuai penamaan domain masing-masing
    file “/etc/bind/rasyid”;              //ganti nama rasyid sesuaikan dengan domain kalian
    type master;
};
zone “110.168.192.in-addr.arpa” {
    file “/etc/bind/db.192”;
    type master;
};
```

Note :

- ***“110.168.192” adalah 3 blok awal ip local***
- ***rasyid.sch.id*** adalah nama domain yang saya pergunakan, sesuaikan nama domain tersebut dengan yang akan kalian gunakan

Copy file **db.local** lalu berikan nama dengan **rasyid** (rasyid wajib sama dengan konfigurasi di dalam **named.conf.local**)

cp /etc/bind/db.local /etc/bind/rasyid hasilnya seperti tampak dibawah ini

```
root@rasyid-server:/# cd /etc/bind
root@rasyid-server:/etc/bind# ls
bind.keys  db.255      named.conf.default-zones  rndc.key
db.0       db.empty    named.conf.local          zones.rfc1918
db.127     db.local    named.conf.options
db.192     named.conf  rasyid
root@rasyid-server:/etc/bind#
```

Edit file **rasyid** lalu sesuaikan konfigurasinya

pico /etc/bind/rasyid

```
  GNU nano 3.2                                  smkn1smr

;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     localhost. root.localhost. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      localhost.
@       IN      A       127.0.0.1
@       IN      AAAA    ::1
```

Ubah menjadi

```
  GNU nano 5.4                                  /etc/bind/rasyid
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     rasyid.sch.id. root.rasyid.sch.id. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      rasyid.sch.id.
@       IN      A       192.168.110.2
www     IN      A       192.168.110.2
@       IN      AAAA    ::1
```

Note :

- **Sesuaikan nama domain (rasyid.sch.id) dan ip (192.168.110.2) yang akan digunakan**

Copy file **db.127** lalu ubah menjadi **db.192** dengan cara berikut

cp /etc/bind/db.127 /etc/bind/db.192

Edit file **db.192**

pico /etc/bind/db.192

```
  GNU nano 3.2                                    db.192

;
; BIND reverse data file for local loopback interface
;
$TTL    604800
@       IN      SOA     localhost. root.localhost. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      localhost.
1.0.0   IN      PTR     localhost.
```

Ubah menjadi

```
  GNU nano 5.4                                    /etc/bind/db.192
;
; BIND reverse data file for local loopback interface
;
$TTL    604800
@       IN      SOA     rasyid.sch.id. root.rasyid.sch.id. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      rasyid.sch.id.
2       IN      PTR     rasyid.sch.id.
```

Note :

- **angka 2 adalah 1 blok terakhir dari ip anda (sesuaikan angka tersebut)**

Pastikan untuk selalu melakukan restart service bind9 setiap kali melakukan perubahan pada konfigurasi

systemctl restart bind9

systemctl status bind9

```
root@debian:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
root@debian:/etc/bind#
```

- **Pengujian**

  Pada versi debian 10, perintah **nslookup** tidak include dari awal (berbeda dengan versi debian sebelumnya yang otomatis terinstall ketika setup debian), untuk itu kita wajib install paket dnsutils, lakukan dengan perintah :

  # apt-get install dnsutils

  # nslookup rasyid.sch.id

```
root@rasyid-server:~# nslookup rasyid.sch.id
Server:         192.168.110.1
Address:        192.168.110.1#53

Non-authoritative answer:
Name:   rasyid.sch.id
Address: 192.168.110.2

root@rasyid-server:~#
```

  Jika terdapat pesan kesalahan, periksa kembali konfigurasi yang telah di buat

KONFIGURASI WEB DAN DATABASE SERVER

9. **Install Aplikasi web server, database server, php dan pendukungnya**

   # apt-get install apache2 mariadb-server php php-curl php-gd libapache2-mod-php php-xml php-mysql php-zip phpmyadmin

   Sebenarnya versi lengkapnya seperti dibawah ini sih

   apache2 apache2-doc apache2-utils libapache2-mod-php php php-common php-gd php-mysql php-imap php-cli php-cgi libapache2-mod-fcgid apache2-suexec-pristine php-pear mcrypt imagemagick libruby libapache2-mod-python php-curl php-intl php-pspell php-sqlite3 php-xsl memcached php-memcache php-imagick php-gettext php-zip php-mbstring memcached libapache2-mod-passenger php-soap php-fpm php-opcache php-apcu

mysql_secure_installation

```
root@smkn1smr:/# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user.  If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

Set root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!


By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
 ... Success!

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.
```

ENTER SAJA

SUPAYA TIDAK BINGUNG
PASTIKAN PASSWORD YG DIISI
SAMA DENGAN LOGIN ROOT

```
Disallow root login remotely? [Y/n] y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
 - Dropping test database...
 ... Success!
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
 ... Success!

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
root@smkn1smr:/# _
```

Login kedalam aplikasi, lalu buat Database dan User yang digunakan nantinya untuk setup aplikasi website kita (dalam praktek kali ini menggunakan wordpress)

mysql -u root -p
MariaDB [(none)]> CREATE DATABASE wordpress;
MariaDB [(none)]> GRANT ALL ON wordpress.* TO 'rasyid'@'localhost' IDENTIFIED BY '123456' WITH GRANT OPTION;
MariaDB [(none)]> FLUSH PRIVILEGES;
MariaDB [(none)]> exit;

Deretan baris perintah itu akan menghasilkan 1 database baru dengan nama wordpress, kemudian memberikan hak akses full kepada user dengan nama rasyid serta password 123456, silahkan sesuaikan untuk penamaannya

Perintah pendukung mariadb
systemctl start mariadb
systemctl restart mariadb
systemctl stop mariadb
systemctl reload mariadb

10. Konfigurasi Virtual Host

Virtual Host ini akan mewakili konfigurasi untuk setiap website yang akan kita buat. Kita dapat mengganti file Virtual Host default yang sudah ada, tapi ada baiknya kita copy saja file tersebut, dan membuat konfigurasi Virtual Host yang baru untuk website kita.

- ✓ Ubah direktori kerja anda menjadi
  # cd /etc/apache2/sites-available/

```
root@smkn1smr:/# cd /etc/apache2
root@smkn1smr:/etc/apache2# ls
apache2.conf    conf-enabled  magic          mods-enabled  sites-available
conf-available  envvars       mods-available  ports.conf    sites-enabled
root@smkn1smr:/etc/apache2# cd sites-available
root@smkn1smr:/etc/apache2/sites-available# ls
000-default.conf  default-ssl.conf
root@smkn1smr:/etc/apache2/sites-available# cp 000-default.conf wordpress.conf
root@smkn1smr:/etc/apache2/sites-available# _
```

- ✓ Copy kan file default dari virtual host yang sudah ada lalu sesuaikan nama dengan yang kita inginkan
  # cp /etc/apache2/sites-available/000-default.conf /etc/apache2/sites-available/wordpress.conf
- ✓ Konfigurasi file yang sudah di copy tadi
  # pico /etc/apache2/sites-available/wordpress.conf
  konfigurasi awal

```
 GNU nano 3.2                                wordpress.conf

<VirtualHost *:80>_
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn
```

Ubah menjadi

```
 GNU nano 5.4                  /etc/apache2/sites-available/wordpress.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerName rasyid.sch.id
        ServerAlias www.rasyid.sch.id
        ServerAdmin rasyid@rasyid.sch.id
        DocumentRoot /var/www/wordpress

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined
```

ServerName rasyid.sch.id **#Silahkan disesuaikan dengan domain masing-masing**
ServerAlias www.rasyid.sch.id **#Silahkan disesuaikan dengan domain masing-masing**
ServerAdmin rasyid@smkn1samarinda.sch.id **#Silahkan disesuaikan dengan domain masing-masing**
DocumentRoot /var/www/wordpress

- ✓ Disable VirtualHost default yang sudah ada, dan aktifkan VirtualHost untuk website utama kita.

  # a2dissite 000-default.conf
  # a2ensite wordpress.conf

```
root@smkn1smr:/etc/apache2/sites-available# ls
000-default.conf  default-ssl.conf  wordpress.conf
root@smkn1smr:/etc/apache2/sites-available# a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@smkn1smr:/etc/apache2/sites-available# a2ensite wordpress.conf
Enabling site wordpress.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@smkn1smr:/etc/apache2/sites-available#
```

11. Konfigurasi Web Direktori

- ✓ Ubah direktori kerja Anda

  # cd /var/www/
- ✓ Buat direktori sesuai dengan nama yang ada dalam konfigurasi virtual host

  # mkdir wordpress
- ✓ Ubah direktori kerja ke folder yang telah di buat

  # cd /var/www/wordpress

  Sehingga menjadi tampil seperti ini
  smkn1smr:/var/www/wordpress#

INSTALASI CMS WORDPRESS

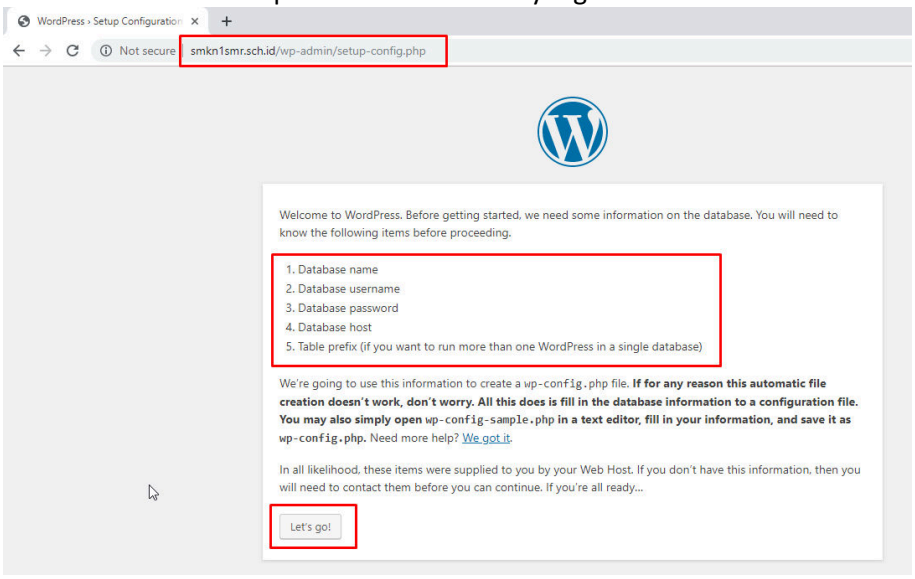
12. Konfigurasi Website CMS Wordpress

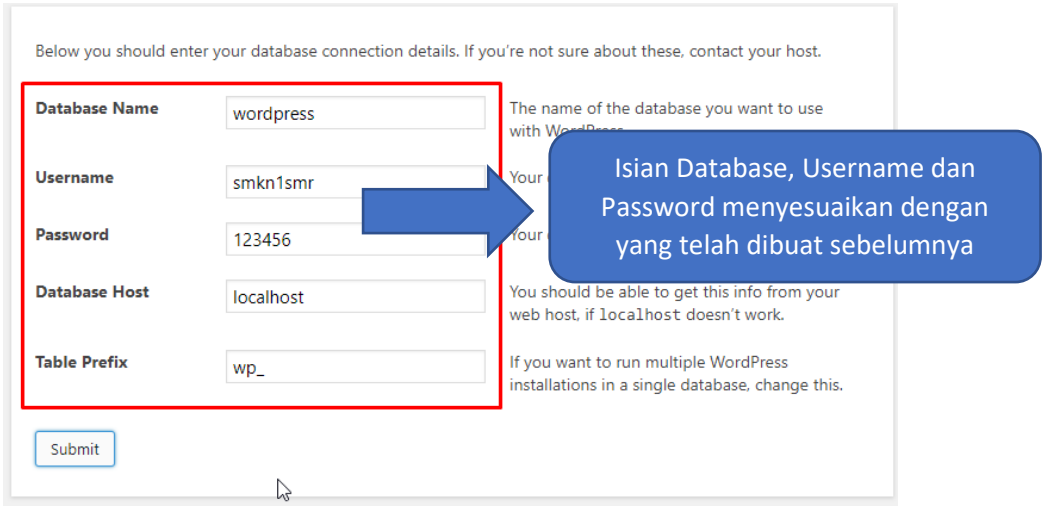
Selanjutnya, tinggal kita masukan saja website kita pada direktori tersebut. Misalnya pada praktikum kali ini akan menggunakan CMS Wordpress, silahkan download pada http lokal yang telah disediakan di **http*://192.168.110.10/debian/wordpress.zip*** Dan letakan pada direktori **wordpress**, kemudian ekstrak. Pastikan Anda mendownload aplikasi **wordpress.zip** di dalam folder aktif **/var/www/wordpress#**

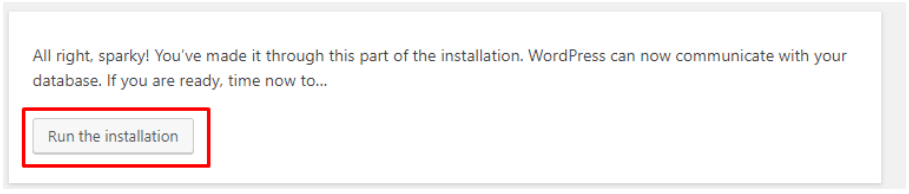
- ✓ Download file wordpress.zip
  /var/www/wordpress# wget http://192.168.110.10/debian/wordpress.zip
  Wget https://id.wordpress.org/latest-id_ID.zip

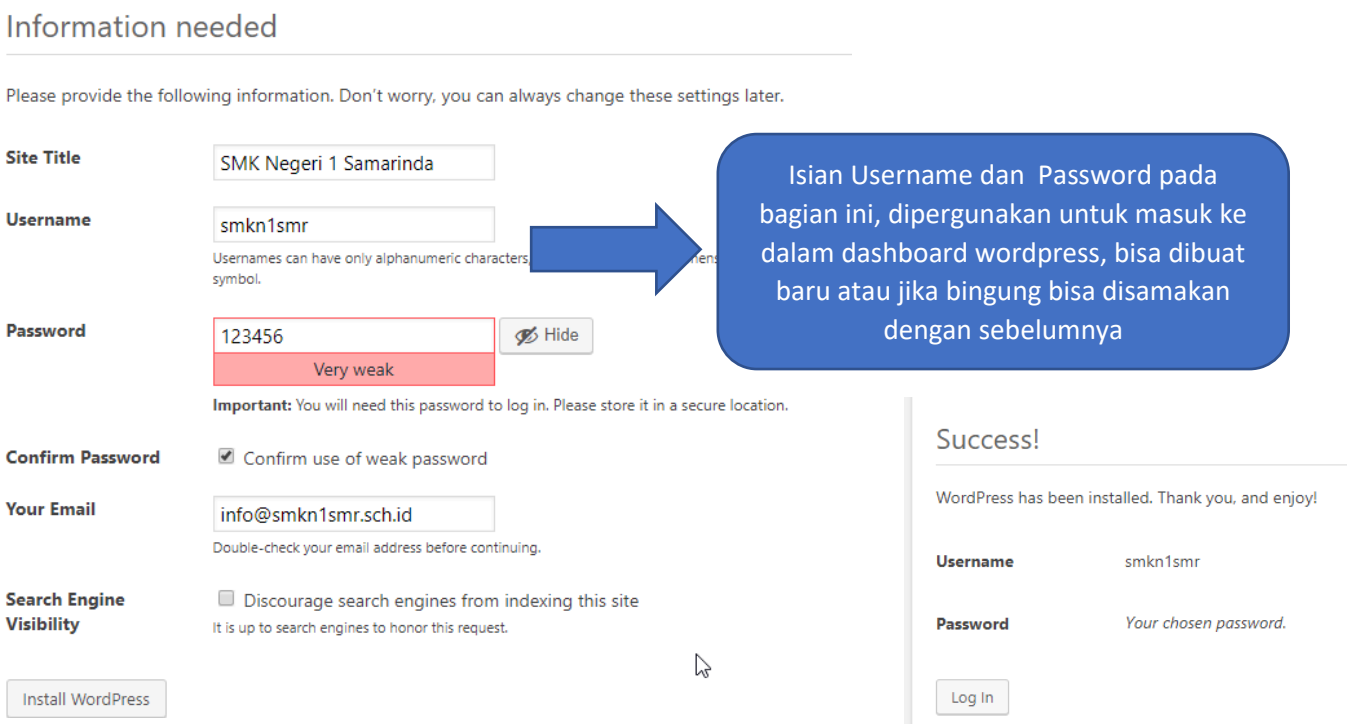
- ✓ Install terlebih dahulu program unzip (jika menggunakan debian 10 biasanya sudah terinstall)
  # apt-get install unzip
- ✓ Ekstrak file wordpress
  /var/www/wordpress# unzip wordpress.zip
  /var/www/wordpress# ls
- ✓ Ubah hak akses direktori tersebut menjadi writeable bagi semua user untuk sementara waktu. Agar installasi Wordpress dapat berjalan lancar.
  var/www/wordpress# chmod 777 ../wordpress/ -R
- ✓ Hal terakhir yang perlu kita lakukan, agar semua konfigurasi yang telah kita setting berjalan, restart aplikasi apache2.
  var/www/wordpress# /etc/init.d/apache2 restart
  atau dengan perintah
  var/www/wordpress# service apache2 restart
  atau dengan perintah
  var/www/wordpress# systemctl restart apache2

Lakukan instalasi wordpress di browser kesayangan anda









Jika status password Very weak, pastikan berikan tanda centang pada **Confirm use of weak password**

Note :
Jika masalah .htaccess tidak dapat terbaca di server debian lokal

edit file /etc/apache2/apache.conf
pada bagian
<Directory /var/www/>
AllowOverride None ubah menjadi AllowOverride All

Selanjutnya aktifkan ModRewrite dengan mengetikkan perintah :
sudo a2enmod rewrite

SETING SHARING INTERNET

13. IP Forwarding

Buka file sysctl.conf yang berada di folder /etc

pico /etc/sysctl.conf

```
# Uncomment the next line to enable packet forwarding for IPv4
#net.ipv4.ip_forward=1
```

Hilangkan tanda pagar sehingga menjadi

```
# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
```

Simpan dan keluar, lalu jalankan perintah **sysctl -p** untuk menerapkan perubahan yang telah di buat

sysctl -p

```
root@smkn1smr:~# sysctl -p
net.ipv4.ip_forward = 1
root@smkn1smr:~#
```

14. IPTABLES NAT (Network Adress Translation)

- ✓ Install aplikasi iptables

  # apt install iptables
- ✓ Jalankan perintah iptables berikut untuk membuka firewall agar dapat dapat di akses internet dari client

  # iptables -t nat -A POSTROUTING -j MASQUERADE

```
root@smkn1smr:~# iptables -t nat -A POSTROUTING -j MASQUERADE
```

- ✓ Konfigurasi IPTABLES tersebut akan selalu hilang ketika setiap kali pc di restart, untuk menghindari hal tersebut kita simpan konfigurasi tersebut dengan cara

  # iptables-save > /etc/network/iptables

```
root@smkn1smr:~# iptables-save > /etc/network/iptables
```

- ✓ Buka file interfaces dan tambahkan scrip berikut dipaling bawah, agar supaya setingan iptables yang kita buat selalu di jalankan setiap kali server di aktifkan

  pre-up iptables-restore < /etc/network/iptables

  Hasil akhir seperti dibawah ini

```
auto enp0s3
iface enp0s3 inet static
address 10.0.2.16/24
gateway 10.0.2.2


auto enp0s8
iface enp0s8 inet static
address 192.168.56.2/24

pre-up iptables-restore < /etc/network/iptables
```

- ✓ Dan pastikan restart service networking nya agar konfigurasi yang dibuat langsung dijalankan

  # service networking restart atau dengan cara **# /etc/init.d/networking restart**

PROXY SERVER

15. Proxy

Proxy server bertugas untuk melakukan cache dan filtering terhadap koneksi internet, aplikasi yang umum digunakan adalah **SQUID**

- ✓ Install apllikasi tersebut dengan cara

  # apt-get install squid

```
root@smkn1smr:~# apt-get install squid
```

- ✓ Lakukan konfigurasi squid pada fle **/etc/squid/squid.conf**

  # pico /etc/squid/squid.conf

```
root@smkn1smr:~# pico /etc/squid/squid.conf
```

Ada begitu banyak konfigurasi yang dapat dilakukan, untuk singkatnya, cukup tambahkan beberapa baris perintah berikut di bagian paling atas dari file squid.conf

acl LAN src 192.168.56.0/24
acl blokir dstdomain "/etc/squid/blokir"
http_access deny blokir
http_access allow LAN
cache_mgr info@rasyid.sch.id
http_port 3128 transparent

Hasil akhir seperti dibawah ini

```
  GNU nano 3.2                    /etc/squid/squid.conf

acl LAN src 192.168.56.0/24
acl blokir dstdomain "/etc/squid/blokir"
http_access deny blokir
http_access allow LAN
cache_mgr rasyid@sch.id
http_port 3128 transparent

#        WELCOME TO SQUID 4.6
#        ----------------------------
#
#        This is the documentation for the Squid configuration
#        This documentation can also be found online at:
#                http://www.squid-cache.org/Doc/config/
```

✓ Buat file blokir yang sudah kita definisikan pada konfigurasi squid, nantinya file ini berisi daftar website yang akan kita blokir

pico /etc/squid/blokir

```
root@smkn1smr:~# pico /etc/squid/blokir
```

Isikan beberapa alamat situs yang akan kita blokir

```
  GNU nano 3.2                                  /etc/squid/blokir

detik.com
www.detik.com
smkn1samarinda.sch.id
www.smkn1samarinda.sch.id
```

✓ Restart service squid

service squid restart atau bisa juga dengan perintah **# /etc/init.d/squid restart**

```
root@smkn1smr:~# service squid restart
root@smkn1smr:~# /etc/init.d/squid restart
[ ok ] Restarting squid (via systemctl): squid.service.
root@smkn1smr:~#
```

Jika masih terdapat eror, cek kembali setingan Anda

✓ Tambahkan kembali scrip berikut ke dalam iptables agar transparent proxy dapat berjalan, lalu simpan

iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to 192.168.120.1:3128

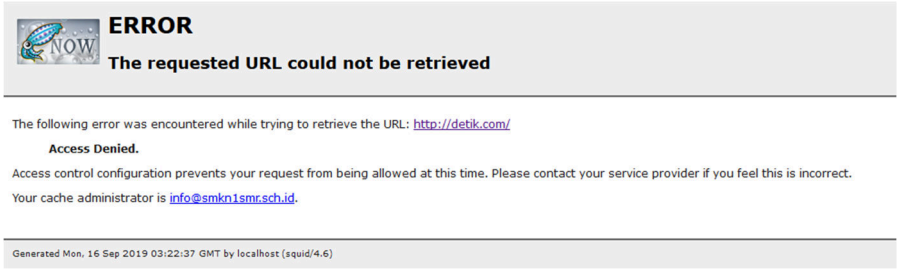
iptables-save > /etc/network/iptables

```
root@smkn1smr:~# iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to 192.168.120.1:3128
root@smkn1smr:~# iptables-save > /etc/network/iptables
root@smkn1smr:~#
```

➢ **Pengujian**

Hubungkan client anda pada ethernet yang ke 2 pada server, dan seting ip ethernet client nya menyesuaikan dengan network pada ether server (gimana cara nya gunakan logika kalian ya)

Lalu cobalah akses ke browser pada alamat yang sudah dimasukkan dalam list blokir, jika tidak dapat di akses, maka setingan Anda sudah benar



DHCP (Dynamic Host Configuration Protocol) SERVER

16. Install Aplikasi DHCP Server

- ✓ Install aplikasi DHCP dengan paket isc-dhcp-server

  # apt-get install isc-dhcp-server

- ✓ Ubah ethernet yang akan dijadikan sebagai penyedia dhcp

  # pico /etc/default/isc-dhcp-server

```
INTERFACESv4="enp0s8"
INTERFACESv6=""
```

  Sesuaikan **enp0s8** dengan nama ethernet Anda yang di gunakan untuk ke client

- ✓ Konfigurasi file dhcpd.conf

  # pico /etc/dhcp/dhcpd.conf

  Sesuaikan dengan setingan di bawah ini.

  Hilangkan tanda **#** pada **authoritative;**

```
# network, the authoritative directive should be uncommented.
authoritative;
```

  Hilangkan tanda **#** mulai baris Subnet hingga **}** lalu sesuaikan konfigurasi nya

```
  GNU nano 3.2                          /etc/dhcp/dhcpd.conf

#  option routers rtr-239-32-1.example.org;
#}

# A slightly different configuration for an internal subnet.
subnet 192.168.56.0 netmask 255.255.255.0 {
  range 192.168.56.5 192.168.56.10;
  option domain-name-servers 192.168.56.2, 10.0.2.2, 8.8.8.8, 8.8.4.4;
  option domain-name "rasyid.sch.id";
  option routers 192.168.56.2;
  option broadcast-address 192.168.56.255;
  default-lease-time 600;
  max-lease-time 7200;
}
```

- ✓ Simpan dan keluar, lalu restart service dhcp

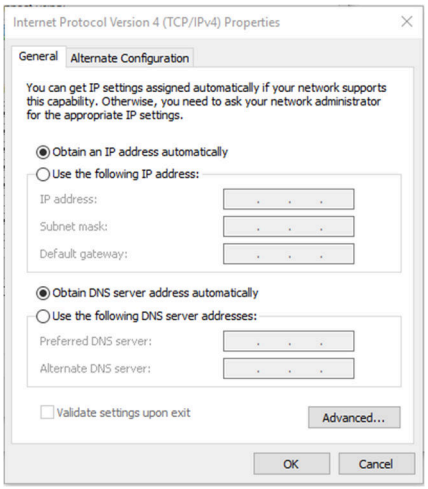
  # /etc/init.d/isc-dhcp-server restart atau dengan cara **# service isc-dhcp-server restart**

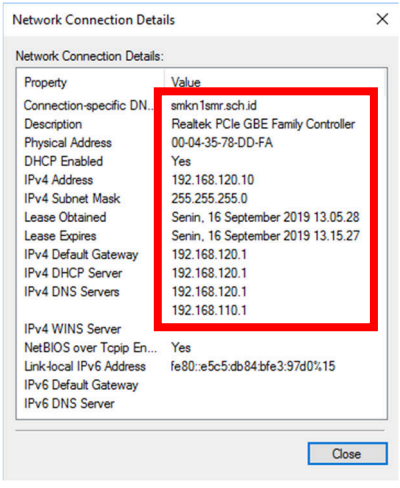
```
root@smkn1smr:~# service isc-dhcp-server restart
root@smkn1smr:~# /etc/init.d/isc-dhcp-server restart
[ ok ] Restarting isc-dhcp-server (via systemctl): isc-dhcp-server.service.
root@smkn1smr:~#
```

  Jika masih terdapat eror, perbaiki kembali setingan Anda

- ➢ **Pengujian**

  Hubungkan pc client Anda dengan server, lalu seting network nya secara dhcp, dan pastikan pc client Anda mendapatkan IP dari server Debian Anda





Cek koneksi jalur internet dari client Anda

```
C:\Users\SMK N 1 Samarinda>tracert kompas.com

Tracing route to kompas.com [202.61.113.35]
over a maximum of 30 hops:

  1     1 ms    <1 ms     1 ms  192.168.120.1
  2     2 ms     2 ms     2 ms  192.168.110.1
  3     2 ms     1 ms     1 ms  10.10.20.1
  4     *        *        *     Request timed out.
  5     6 ms     7 ms     5 ms  10.100.16.1
  6     6 ms     6 ms     6 ms  180.252.1.149
  7    32 ms    31 ms    31 ms  telkom.openixp.net [218.100.36.56]
  8    32 ms    31 ms     *     tengiga-0-1.openixp.net [218.100.36.2]
  9    34 ms    41 ms    32 ms  kompas.openixp.net [218.100.36.67]
 10    32 ms    32 ms    32 ms  regional.kompas.com [202.61.113.35]
 11    32 ms    31 ms    32 ms  regional.kompas.com [202.61.113.35]

Trace complete.
```

Perhatikan bahwa akses client Anda itu melalui IP dari ethernet Lan Server, lalu di teruskan ke IP ether internet server Anda

SETING FILE TRANSFER PROTOCOL (FTP)

17. Install Aplikasi FTP

FTP Server merupakan layanan berbagi file client dan server, dapat melakukan upload dan download

✓ Install paket ftp

apt-get install proftpd

```
root@smkn1smr-server:/# apt-get install proftpd
```

✓ Buat folder untuk meletakkan semua file dan data pada ftp server, kemudian tambahkan hak akses untuk user ftp pada folder tersebut

cd /home/smkn1smr **// smkn1smr sesuikan dengan akun masing-masing**

mkdir share **// PERINTAH MEMBUAT FOLDER**

userdel ftp **// PERINTAH MENGHAPUS USER FTP (dibuat otomatis ketika install proftpd)**

useradd -d /share/ ftp **// PERINTAH MEMBUAT USER FTP**

passwd ftp **// PERINTAH MEMBUAT PASSWORD UNTUK USER FTP**

Isikan password misal 123456

```
root@smkn1smr-server:/home/smkn1smr# mkdir share
root@smkn1smr-server:/home/smkn1smr# userdel ftp
root@smkn1smr-server:/home/smkn1smr# useradd -d /share/ ftp
root@smkn1smr-server:/home/smkn1smr# passwd ftp
New password:
Retype new password:
passwd: password updated successfully
root@smkn1smr-server:/home/smkn1smr# _
```

✓ Ubah hak akses permission pada folder ftp share yang telah kita buat

chmod 777 -R /home/smkn1smr/share

```
root@smkn1smr-server:/home/smkn1smr# chmod 777 -R /home/smkn1smr/share
root@smkn1smr-server:/home/smkn1smr# _
```

✓ User Authentication Login

Edit dan tambahkan script berikut di ***baris paling bawah*** dari file **proftpd.conf**

pico /etc/proftpd/proftpd.conf

```
root@smkn1smr-server:/# pico /etc/proftpd/proftpd.conf_
```

<Anonymous /home/smkn1smr/share/>
User ftp
</Anonymous>

```
# Include other custom configuration files
Include /etc/proftpd/conf.d/

<Anonymous /home/smkn1smr/share/>
User    ftp
</Anonymous>
```

✓ Restart service ftp (silahkan pilih salah satu caranya)

/etc/init.d/proftpd restart
systemctl restart proftpd **//perintah restart service proftpd**
systemctl status proftpd **//perintah melihat status service proftpd**

```
root@smkn1smr-server:/# /etc/init.d/proftpd restart
[ ok ] Restarting proftpd (via systemctl): proftpd.service.
root@smkn1smr-server:/# _
```

➢ **Pengujian**

Buka browser atau explorer kesayangan Anda lalu akses dengan alamat ftp://ipaddress atau dengan ftp://domain

NOTE :

Perbedaan perintah useradd dan adduser

- useradd akan menciptakan user tanpa diminta untuk mengeset password user tersebut. Sedangkan adduser akan menciptakan user dengan mengeset password user tersebut sekaligus.
- useradd akan menciptakan user tanpa ada sebuah home directory. Sedangkan adduser akan menciptakan user sekaligus dengan home directory.
- useradd akan menciptakan user tanpa memberikan informasi lebih lengkap tentang user. Sedangkan adduser akan menciptakan user sekaligus dapat memberikan keterangan lebih lengkap tentang user.

SETING MAIL SERVER DAN WEB MAIL SERVER

18. Postfix Mail Server

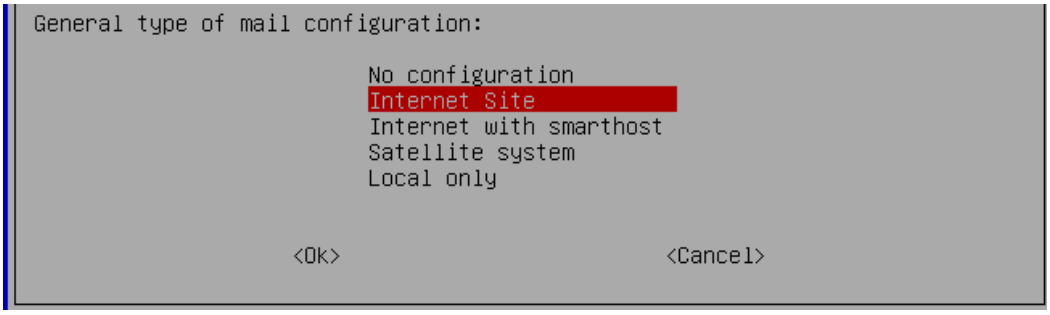
✓ Install aplikasi Postfix

apt-get install postfix

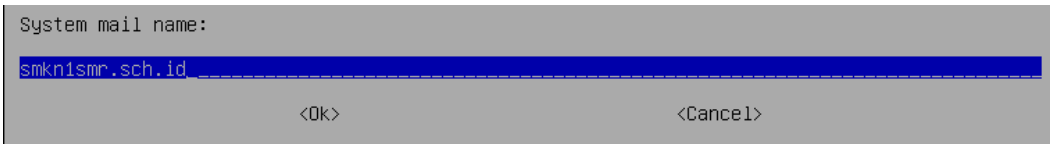
```
root@smkn1smr-server:/# apt-get install postfix_
```

Selama proses install akan muncul menu popup, sesuaikan dengan konfigurasi yang akan kita pakai

- **General Type of Mail Configuration : Internet site**



- **System Mail Name : rasyid.sch.id**



✓ Konfigurasi Postfix

Backup terlebih dahulu konfigurasi asli file postfix

cp /etc/postfix/main.cf /etc/postfix/main.backup

```
root@smkn1smr-server:/# cp /etc/postfix/main.cf /etc/postfix/main.backup
root@smkn1smr-server:/#
```

✓ Edit file main.cf

pico /etc/postfix/main.cf

```
root@smkn1smr-server:/# pico /etc/postfix/main.cf_
```

Ganti **myhostname**, lalu tambahkan **mydomain**

```
myhostname = mail.smkn1smr.sch.id
mydomain = smkn1smr.sch.id
```

Ubah **mynetworks** variables dengan konfigurasi yang kita inginkan

Konfigurasi awal

```
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
```

Ubah menjadi

```
mynetworks = 127.0.0.0/8 192.168.110.0/24_
```

Berikan tanda pagar pada **inet_protocols = all** dan tambahkan baris **inet_protocols = ipv4** serta **home_mailbox = Maildir/**

```
recipient_delimiter = +
inet_interfaces = all
#inet_protocols = all
inet_protocols = ipv4
home_mailbox = Maildir/
```

Tambahkan setingan SMTP berikut di paling bawah

SMTP-Auth settings

smtpd_sasl_type = dovecot

smtpd_sasl_path = private/auth

smtpd_sasl_auth_enable = yes

smtpd_sasl_security_options = noanonymous

smtpd_sasl_local_domain = $myhostname

smtpd_recipient_restrictions = permit_mynetworks,permit_auth_destination,permit_sasl_authenticated,reject

✓ Jalankan postconf -n untuk mengecek konfigurasi pada postfix

postconf -n

```
root@smkn1smr-server:/# postconf -n
```

✓ Setelah selesai konfigurasi, restart Postfix

/etc/init.d/postfix restart atau dengan **# systemctl restart postfix**

```
root@smkn1smr-server:/# /etc/init.d/postfix restart
[ ok ] Restarting postfix (via systemctl): postfix.service.
root@smkn1smr-server:/# systemctl restart postfix
```

✓ Cek status postfix

systemctl status postfix

```
root@smkn1smr-server:/# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (exited) since Tue 2019-09-17 11:52:53 WITA; 2min 4s ago
  Process: 3138 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
 Main PID: 3138 (code=exited, status=0/SUCCESS)

Sep 17 11:52:53 smkn1smr-server systemd[1]: Starting Postfix Mail Transport Agent...
Sep 17 11:52:53 smkn1smr-server systemd[1]: Started Postfix Mail Transport Agent.
root@smkn1smr-server:/#
```

✓ Cek apakah port 25 sudah listen

Jika netstat tidak dapat dijalankan secara langsung, install terlebih dahulu aplikasi net-tools

apt-get install net-tools

netstat -tlpn

```
root@smkn1smr-server:/# netstat -tlpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN      455/mysqld
tcp        0      0 192.168.120.1:53        0.0.0.0:*               LISTEN      437/named
tcp        0      0 192.168.110.3:53        0.0.0.0:*               LISTEN      437/named
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN      437/named
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      456/sshd
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN      3137/master
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN      437/named
tcp6       0      0 :::80                   :::*                    LISTEN      635/apache2
tcp6       0      0 :::21                   :::*                    LISTEN      1234/proftpd: (acce
tcp6       0      0 :::53                   :::*                    LISTEN      437/named
tcp6       0      0 :::22                   :::*                    LISTEN      456/sshd
tcp6       0      0 :::3128                 :::*                    LISTEN      625/(squid-1)
tcp6       0      0 ::1:953                 :::*                    LISTEN      437/named
root@smkn1smr-server:/#
```

✓ Test Postfix Mail Server dalam Debian

Install paket Mailutils, lalu ikuti langkah berikut nya, lakukan perintah di bawah ini dari posisi folder root **cd /root**

apt-get install mailutils

echo "mail body"| mail -s "test mail" root

mailq

mail

ls Maildir/

ls Maildir/new/

cat Maildir/new/[TEKAN TOMBOL TAB] **//[TAB] = MENEKAN TOMBOL TAB KEYBOARD**

```
root@smkn1smr-server:~#
root@smkn1smr-server:~# echo "mail body" | mail -s "test mail" root
root@smkn1smr-server:~# mailq
Mail queue is empty
root@smkn1smr-server:~# mail
No mail for root
root@smkn1smr-server:~# ls Maildir/
cur  new  tmp
root@smkn1smr-server:~# ls Maildir/new/
1568702716.V801I1016feM905944.smkn1smr-server
root@smkn1smr-server:~# cat Maildir/new/1568702716.V801I1016feM905944.smkn1smr-server
Return-Path: <root@smkn1smr.sch.id>
X-Original-To: root@smkn1smr.sch.id
Delivered-To: root@smkn1smr.sch.id
Received: by mail.smkn1smr.sch.id (Postfix, from userid 0)
        id CDE7F61271; Tue, 17 Sep 2019 14:45:16 +0800 (WITA)
Subject: test mail
To: <root@smkn1smr.sch.id>
X-Mailer: mail (GNU Mailutils 3.5)
Message-Id: <20190917064516.CDE7F61271@mail.smkn1smr.sch.id>
Date: Tue, 17 Sep 2019 14:45:16 +0800 (WITA)
From: root <root@smkn1smr.sch.id>

mail body
root@smkn1smr-server:~#
```

19. Install dan Konfigurasi Dovecot

The mail delivery agent that we'll be using in this guide to deliver e-mail messages to a local recipient's mailboxes is Dovecot IMAP. IMAP is a protocol which runs on 143 and 993 (SSL) ports, which is responsible reading, deleting or moving mails across multiple email clients.

The IMAP protocol also uses synchronization in order to assure that a copy of each message is saved on the server and allows users to create multiple directories on the server and move mails to this directories in order to sort the e-mails.

This is not the case with **POP3** protocol. **POP3** protocol won't allow users to create multiple directories on the server to sort your mail. You only have the inbox folder to manage mail.

Nah loh, artikan sendiri yaa

- ✓ Install Dovecot core server dan Dovecot IMAP

  # apt install dovecot-core dovecot-imapd

  ```
  root@smkn1smr-server:~# apt-get install dovecot-core dovecot-imapd
  ```

- ✓ Edit file dovecot.conf

  # pico /etc/dovecot/dovecot.conf

  Cari #listen = *, :: dan hilangkan tanda pagar

  `#listen = *, ::` menjadi `listen = *, ::`

- ✓ Edit file 10-auth.conf

  # pico /etc/dovecot/conf.d/10-auth.conf

  ```
  root@smkn1smr-server:~# pico /etc/dovecot/conf.d/10-auth.conf
  ```

  Cari dan ubah hingga menjadi seperti yang dibawah ini

  ```
  #disable_plaintext_auth = yes
  disable_plaintext_auth = no
  auth_mechanisms = plain login
  ```

- ✓ Edit file **10-mail.conf**

  # pico /etc/dovecot/conf.d/10-mail.conf

  ```
  root@smkn1smr-server:~# pico /etc/dovecot/conf.d/10-mail.conf
  ```

  Tambahkan scrip berikut

mail_location = maildir:~/Maildir

```
#mail_location = mbox:~/mail:INBOX=/var/mail/%u
mail_location = maildir:~/Maildir
```

✓ Edit file **10-master.conf**

pico /etc/dovecot/conf.d/10-master.conf

```
root@smkn1smr-server:~# pico /etc/dovecot/conf.d/10-master.conf
```

Hilangkan tanda pagar nya

```
# Postfix smtp-auth
#unix_listener /var/spool/postfix/private/auth {
#  mode = 0666
#}
```

Ubah menjadi

```
 #Postfix smtp-auth
 unix_listener /var/spool/postfix/private/auth {
   mode = 0666
   user = postfix
  group = postfix
 }
```

✓ Restart service dovecot

systemctl restart dovecot.service

systemctl status dovecot.service

```
root@smkn1smr-server:~# systemctl restart dovecot.service
root@smkn1smr-server:~# systemctl status dovecot.service
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2019-09-17 16:00:02 WITA; 5s ago
     Docs: man:dovecot(1)
           http://wiki2.dovecot.org/
 Main PID: 5594 (dovecot)
    Tasks: 4 (limit: 4788)
   Memory: 2.1M
   CGroup: /system.slice/dovecot.service
           ├─5594 /usr/sbin/dovecot -F
           ├─5595 dovecot/anvil
           ├─5596 dovecot/log
           └─5597 dovecot/config

Sep 17 16:00:02 smkn1smr-server systemd[1]: Started Dovecot IMAP/POP3 email server.
Sep 17 16:00:02 smkn1smr-server dovecot[5594]: master: Dovecot v2.3.4.1 (f79e8e7e4) starting up for

root@smkn1smr-server:~# _
```

netstat -tlpn

```
root@smkn1smr-server:~# netstat -tlpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:143             0.0.0.0:*               LISTEN      5594/dovecot
tcp        0      0 192.168.120.1:53        0.0.0.0:*               LISTEN      404/named
tcp        0      0 192.168.110.3:53        0.0.0.0:*               LISTEN      404/named
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN      404/named
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      417/sshd
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN      5570/master
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN      404/named
tcp        0      0 0.0.0.0:993             0.0.0.0:*               LISTEN      5594/dovecot
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN      455/mysqld
tcp6       0      0 :::143                  :::*                    LISTEN      5594/dovecot
tcp6       0      0 :::80                   :::*                    LISTEN      582/apache2
tcp6       0      0 :::21                   :::*                    LISTEN      755/proftpd: (accep
tcp6       0      0 :::53                   :::*                    LISTEN      404/named
tcp6       0      0 :::22                   :::*                    LISTEN      417/sshd
tcp6       0      0 :::3128                 :::*                    LISTEN      563/(squid-1)
tcp6       0      0 ::1:953                 :::*                    LISTEN      404/named
tcp6       0      0 :::993                  :::*                    LISTEN      5594/dovecot
root@smkn1smr-server:~# _
```

20. Buat user tes mail

✓ Menambahkan user baru dengan nama **siswa1** dan password **siswa1** dan testing kirim email

adduser siswa1

```
root@smkn1smr-server:/# adduser siswa1
Adding user `siswa1' ...
Adding new group `siswa1' (1002) ...
Adding new user `siswa1' (1002) with group `siswa1' ...
Creating home directory `/home/siswa1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for siswa1
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@smkn1smr-server:/#
```

nc localhost 25
ehlo localhost
mail from: root
rcpt to: siswa1
data
subject: test email
Isikan pesan email
.
quit

prosesnya seperti dibawah ini

```
root@smkn1smr-server:/# nc localhost 25
220 mail.smkn1smr.sch.id ESMTP Postfix (Debian/GNU)
ehlo localhost
250-mail.smkn1smr.sch.id
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-AUTH PLAIN
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
mail from: root
250 2.1.0 Ok
rcpt to: siswa1
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
subject: test email
kirim kirim email nah
.
250 2.0.0 Ok: queued as 1C92961358
quit
221 2.0.0 Bye
root@smkn1smr-server:/# _
```

✓ Cek email masuk

ls /home/siswa1/Maildir/new/
cat /home/siswa1/Maildir/new/TAB

```
root@smkn1smr-server:/# ls /home/siswa1/Maildir/new
1568763533.V801Ic2842M717033.smkn1smr-server
root@smkn1smr-server:/# cat /home/siswa1/Maildir/new/1568763533.V801Ic2842M717033.smkn1smr-server
Return-Path: <root@smkn1smr.sch.id>
X-Original-To: siswa1
Delivered-To: siswa1@smkn1smr.sch.id
Received: from localhost (smkn1smr.sch.id [192.168.110.3])
        by mail.smkn1smr.sch.id (Postfix) with ESMTP id 1C92961358
        for <siswa1>; Wed, 18 Sep 2019 07:36:28 +0800 (WITA)
subject: test email
Message-Id: <20190917233700.1C92961358@mail.smkn1smr.sch.id>
Date: Wed, 18 Sep 2019 07:36:28 +0800 (WITA)
From: root@smkn1smr.sch.id

kirim kirim email nah
root@smkn1smr-server:/#
```

SETING WEB MAIL SERVER

21. Konfigurasi Virtual Host untuk Mail Server

- ✓ Buat Virtual host

  Copy kan 000-default.conf dan ubah namanya menjadi mail.conf

  # cp /etc/apache2/sites-available/000-default.conf /etc/apache2/sites-available/mail.conf
- ✓ Edit mail.conf

  # pico /etc/apache2/sites-available/mail.conf

  Sesuaikan isi nya seperti berikut

```
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerName mail.smkn1smr.sch.id
        ServerAdmin info@smkn1smr.sch.id
        DocumentRoot /var/www/mail
```

- ✓ Aktifkan virtual host mail.conf

  # a2ensite mail.conf
- ✓ Buat folder mail

  # mkdir /var/www/mail
- ✓ Ubah direktori kerja ke folder yang baru di buat

  # cd /var/www/mail

22. Install Aplikasi web mail Rainloop

Jalankan perintah berikut (WAJIB DI LAKUKAN DI FOLDER AKTIF **MAIL**)

- ✓ Install Rainloop Mail

  # apt-get install curl

  # curl -sL https://repository.rainloop.net/installer.php | php

JIKA INSTALL ONLINE GAGAL, LAKUKAN LANGKAH DI BAWAH INI, KALO TIDAK GAGAL JANGAN WAL LAAI PAHAM LAAH

wget http://192.168.110.10/debian/rainloop.zip

- ✓ Ekstrak file rainloop.zip yang sudah di download

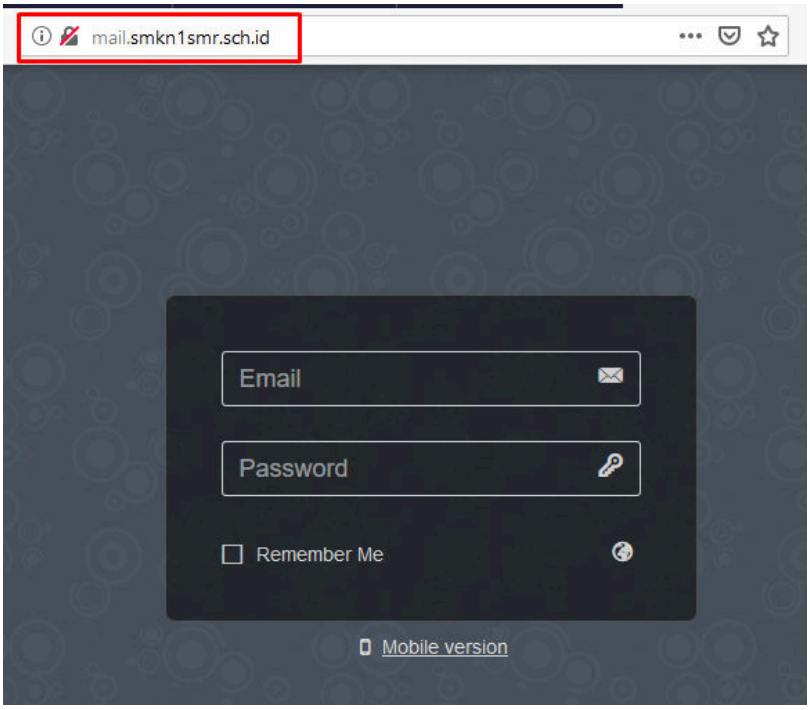
  # unzip rainloop.zip
- ✓ Buat folder mail menjadi writeable

  # chmod 777 ../mail/ -R
- ✓ Restart service apache

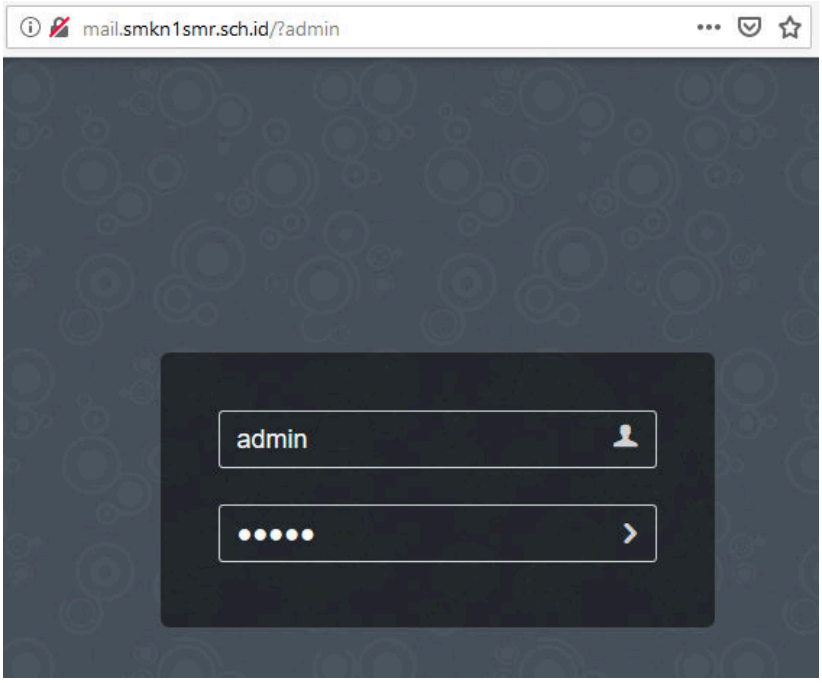
  # service apache2 restart

23. Konfigurasi Rainloop

✓ Akses web mail menggunakan browser kesayangan Anda, arahkan ke alamat **mail.smkn1smr.sch.id**

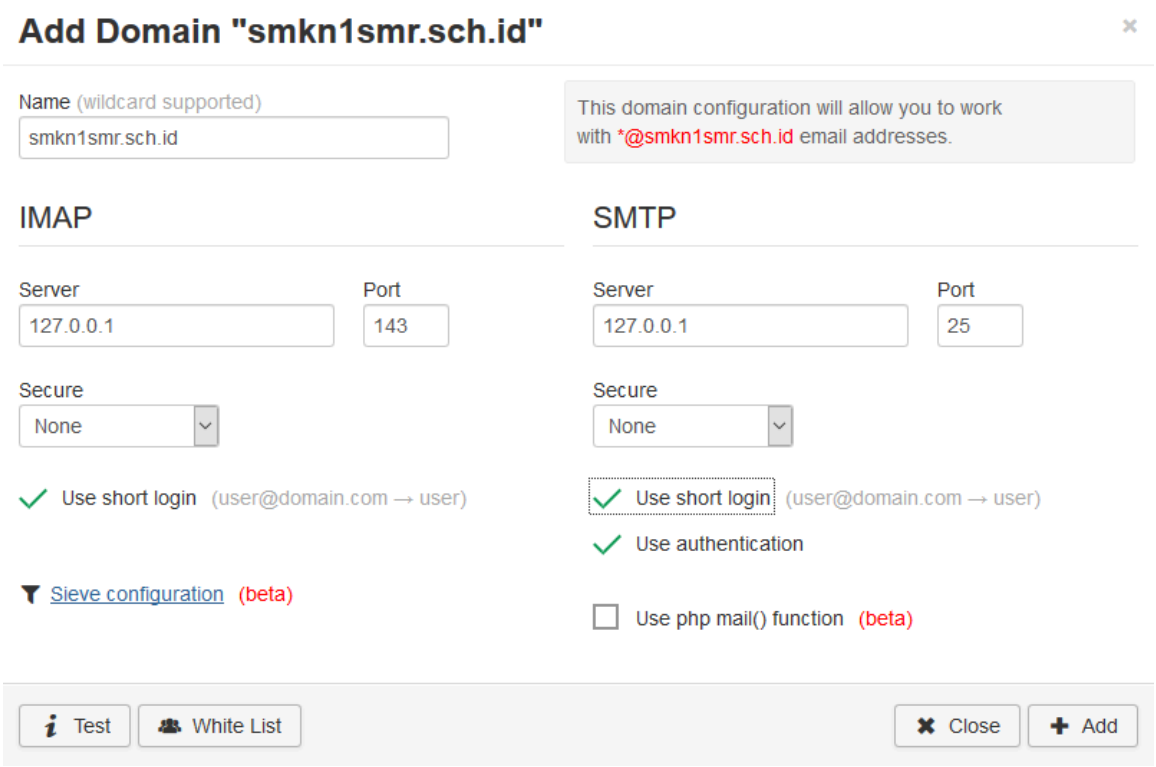


✓ Konfigurasi rainloop dengan mengetikkan url **mail.smkn1smr.sch.id/?admin** Isikan **user = admin pass = 12345**



Sesuaikan seperti berikut ini

Domain → Add Domain



Buatkan user mail yang baru, misal siswa2, kemudian lakukan pengujian dengan cara saling mengirim email menggunakan user mail yang telah di buat

NTP (Network Time Protocol) Server

24. Install Aplikasi

- ✓ Install terlebih dahulu paket ntp (server) dan ntpdate (client)

  # apt-get install ntp ntpdate

- ✓ Konfigurasi

  # pico /etc/ntp.conf

  Edit sehingga menjadi seperti berikut

```
# pool 0.debian.pool.ntp.org iburst
# pool 1.debian.pool.ntp.org iburst
# pool 2.debian.pool.ntp.org iburst
# pool 3.debian.pool.ntp.org iburst
server 127.127.1.0
fudge 127.127.1.0 stratum 1
```

```
# Local users may interrogate the ntp server more closely.
restrict 127.0.0.1
restrict ::1
restrict 192.168.120.0 mask 255.255.255.0 nomodify notrap nopeer
# Needed for adding pool entries
restrict source notrap nomodify noquery
```

- ✓ Ubah tanggal bulan tahun serta jam server Anda

  # date --set yyy-mm-dd

  # date --set jam:menit:detik

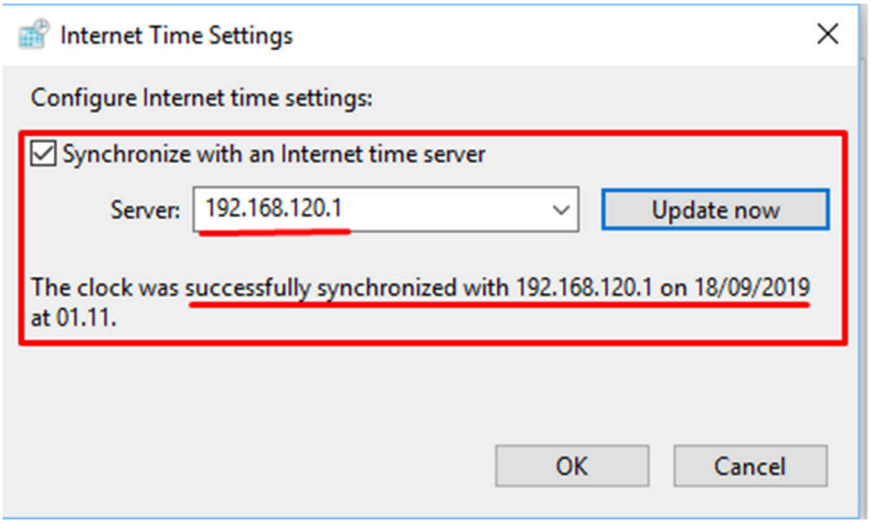
- ✓ Restart ntp server Anda

  # /etc/init.d/ntp restart

  # ntpq -p

- ➢ Pengujian

  Ubah setingan Adjust date/time pada Client windows Anda



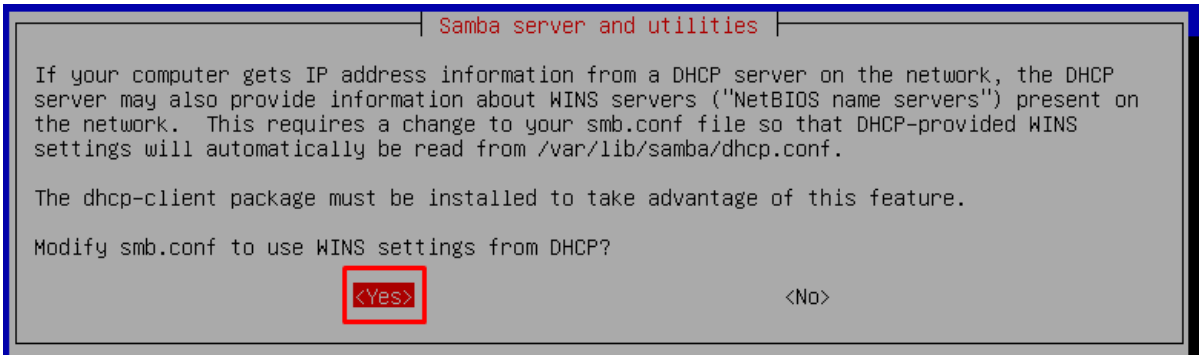
Jika berhasil setingan waktu pada pc Anda akan mengikuti server

FILE SERVER DENGAN SAMBA

25. Install Aplikasi

- ✓ Install terlebih dahulu paket samba

  # apt-get install samba



- ✓ Buat folder yang akan di akses dalam share via jaringan, dan ubah pula metode penulisan folder tersebut

  # cd /home/smkn1smr

  # mkdir samba

  # cd samba

  # chmod 777 ../samba

- ✓ Konfigurasi Samba

  # pico /etc/samba/smb.conf

  Tambahkan baris perintah berikut di akhir file

  [share]

  path = /home/smkn1smr/samba

  browseable = yes

  writeable = yes

  read only = no

  public = yes

- ✓ Restart samba

  # systemctl restart smbd atau dengan perintah **# /etc/init.d/smbd restart**

- ➢ Pengujian

  Akses dari pc client anda lewat windows/file explorer dengan cara mengetikkan pada bagian **url \\alamatIP**

[Nama Folder]
path = /home/rafi/samba/public
browseable = yes
writeable = yes
security = user
read only = no
valid user = rafi
guest ok = no

untuk memasukkan/mengganti password user yg mengakses samba
lakukan perintah ini dari terminal
smbpasswd -a rafi (sesuaikan sama valid user rafi)

SWAT (Samba Web Administration Tool)
26.

VOIP dengan Asterisk dan Zoiper

27. Install Aplikasi Asterisk

- ✓ Install aplikasi asterisk

  # apt-get install asterisk

- ✓ Konfigurasi asterisk

  # pico /etc/asterisk/sip.conf

  Tambahkan konfigurasi berikut pada baris paling bawah

```
[1234]                          # ini menjadi nomor telepon yang akan dituju
context=smkn1samarinda
type=friend
username=1234                   # digunakan untuk login aplikasi voip user@alamat ip
secret=rahasia                  # password login aplikasi
host=dynamic
videosupport=yes

[5678]
context=smkn1samarinda
type=friend
username=5678
secret=rahasia
host=dynamic
videosupport=yes
```

- ✓ Konfigurasi extension asterisk

  # pico /etc/asterisk/extensions.conf

  Tambahkan ke baris paling bawah

```
[smkn1samarinda]                # wajib sama dengan isian context pada file sip.conf
exten =>1234,1,Dial(SIP/1234)
exten =>5678,1,Dial(SIP/5678)
```

Setingan sip.conf

```
[general]
port=5060
bindaddr=0.0.0.0
context=others
allowoverlap=no
srvlookup=yes
videosupport=yes

[100]
type=friend
username=100
context=myphones
secret=1234
```

```
host=dynamic
videosupport=yes
disallow=all
allow=ulaw
allow=alaw
canreinvite=no
allow=gsm
allow=h263p
allow=h264
allow=vp8

[150]
type=friend
username=150
context=myphones
secret=1234
host=dynamic
videosupport=yes
disallow=all
allow=ulaw
allow=alaw
canreinvite=no
allow=gsm
allow=h263p
allow=h264
allow=vp8
```

Setingan extension.conf

```
[general]
static=yes
writeprotect=no
clearglobalvars=no

[others]
[myphones]
exten => 100,1,Dial(SIP/100)
exten => 150,1,Dial(SIP/150)
```

- ✓ Download Aplikasi Zoiper (free dan berbayar, desktop dan android) / Jitsi Meet / Linphone / MizuDroid (tes ok) / grand stream wave pada playstore jika menggunakan Smartphone Android, atau dapat juga download zoiper for windows jika menggunakan pc atau laptop

Restart Service Asterisk

Systemctl restart asterisk

Aplikasi WEB Panel

28. Install Aplikasi Webmin

- ✓ **Install Dependencies webmin**

  # apt-get install perl libnet-ssleay-perl openssl libauthen-pam-perl libpam-runtime libio-pty-perl apt-show-versions python

- ✓ **Installing on Debian**

  Download DEB version

  # wget http://prdownloads.sourceforge.net/webadmin/webmin_1.941_all.deb

  atau download ke server FTP lokal

  # wget ftp://192.168.110.19/os/debian/webmin.deb

  Kemudian jalankan perintah berikut

  # dpkg --install webmin_1.941_all.deb

  Perhatikan nama file yang akan di install, wajib sama dengan nama file yang didownload

  Hasil instalasi otomatis akan akan berada di path /usr/share/webmin

  Jika masih gagal, lakukan perintah di bawah ini

  # apt --fix-broken install

- ✓ **Akses Webmin**

  Secara default Administrasi webmin menggunakan user **root** dan password **root** server anda, silahkan login dengan menggunakan browser pada URL http://localhost:10000/ ganti localhost dengan domain atau ip anda masing-masing.

Jika Debian mengalami masalah pada instalasi dependencies, silahkan install beberapa aplikasi berikut ini

- ✓ **Using the Webmin APT repository**

  Edit file **/etc/apt/sources.list** anda lalu tambahkan baris berikut

  deb https://download.webmin.com/download/repository sarge contrib

  simpan dan keluar

  jangan lupa update repo menggunakan **apt-get update**

  Install my GPG key with which the repository is signed, dengan perintah

  # cd /root

  # wget https://download.webmin.com/jcameron-key.asc

  # apt-key add jcameron-key.asc

  # apt-get install apt-transport-https

  # apt-get update

  # apt-get install webmin

  Semua permasalahan dependencies seharusnya terselesaikan secara otomatis.

Mengubah tampilan SSH Login

29. Mengubah tampilan login ssh

- ✓ Step1#: Login SSH

  Masuk ke server menggunakan ssh

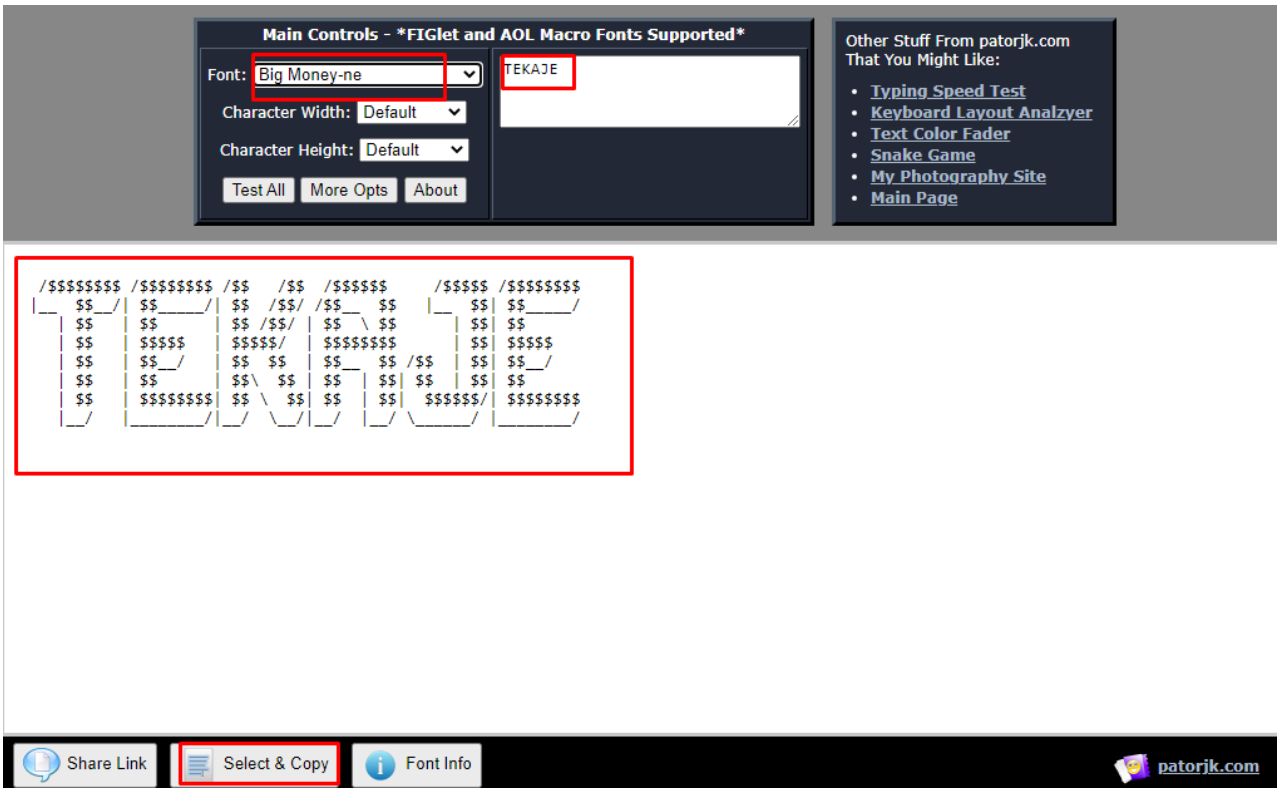
- ✓ Step2#: Edit file motd

  ketik command berikut

  # pico /etc/motd

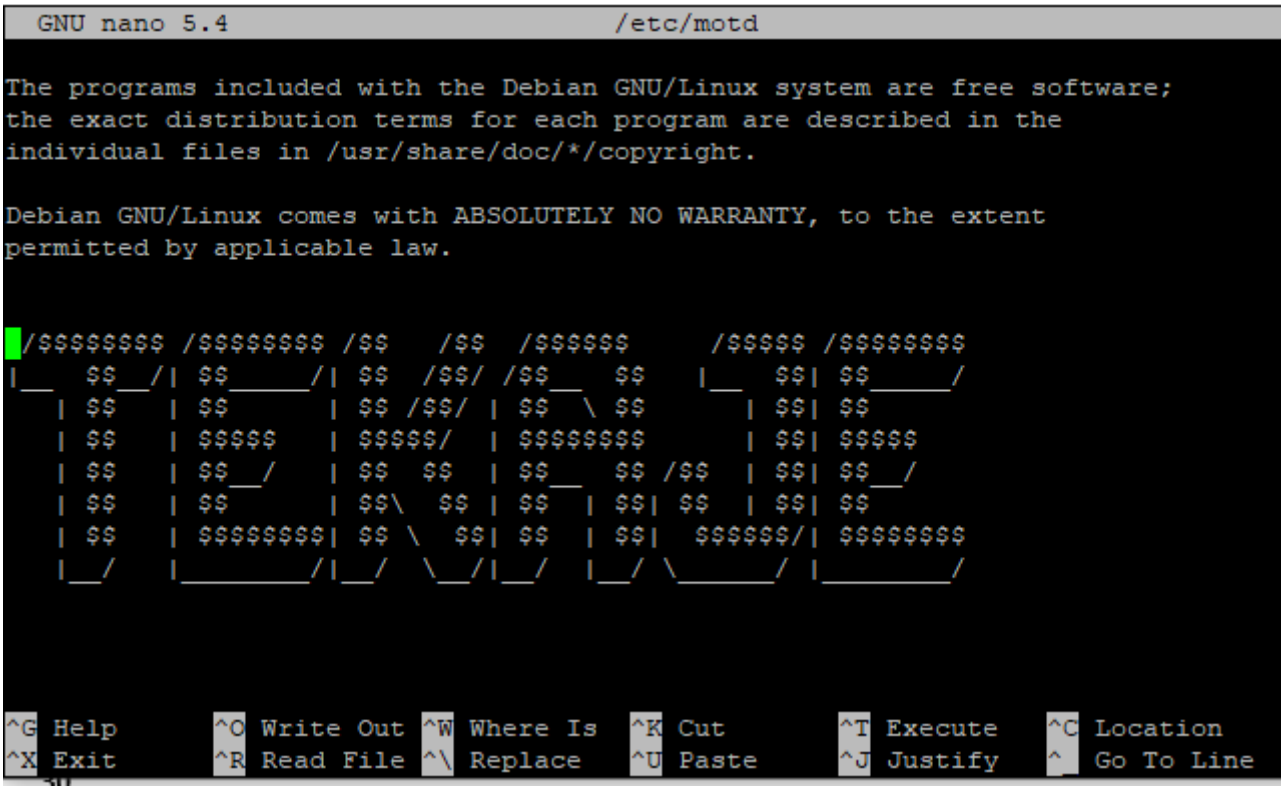
✓ Step3#: Modifikasi Tampilan

Untuk mempermudah silahkan akses https://patorjk.com/software/taag/ bisa juga menggunakan https://www.asciiart.eu/ atau www.text-image.com

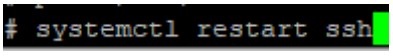


isi kata yang ingin dimasukkan – lalu paste di editor

✓ Hasil editor seperti berikut



✓ Lakukan restart service SSH



✓

30.

Langkah konfig:

1. Instal Virtual Box
2. Install laragon
3. buat folder debian pada c:\laragon\www\
4. buat folder disk1 disk2 disk3 di dalam folder debian
5. ekstrak file iso (bisa gunakan 7zip (ada dalam aplikasi yg di copy) kedalam masing2 disk1 disk2 disk3
6. copykan cumulus dan wordpress ke dalam folder debian
7. buka browser lalu jalankan http://localhost/debian

selamat mencoba

Posisi nya seperti ini

